

A closer look at what/how we can learn from
computational modelling of phonotactics

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Abstract

Computational modelling has been extensively used in recent decades in attempts to understand and probe claims about the nature of phonotactic grammars. In this paper, I argue that much more care is needed in inferring aspects of the underlying competence system from model performance, even when compared against human acceptability judgements. I specifically focus on the issue of gradience and show using the Hayes & Wilson (Maximum-entropy based) Phonotactic Model that despite a good fit to acceptability judgements across three prior experimental results focussed on noncewords targetting a variation in word-initial consonant sequences, the model performance does not stem from gradience in the predictions, or gradience in constraints weights, or gradience in the training data (*i.e.*, type frequency information). I further show that a minimal (categorical) phonotactic model does at least as well on all three experimental results. The latter model also reinforces the distinction between the idea of gradience due to the grammar (extensional gradience) and the grammar itself being gradient (intensional gradience). A more general point made is that, as with all theory testing using experiments, it is difficult to learn about the underlying grammatical system from computational models if the models involve auxiliary hypotheses that themselves are not carefully explored and controlled for.

1 Introduction

Native speakers seem to have intuitions about what is or isn't a reasonable nonsense word of their language. As Chomsky and Halle (1968) note, while /blik/ and /bnik/ are both nonsense words of English, English speakers generally prefer /blik/ to /bnik/. There have been many experimental studies that have explored such word acceptability judgments (Albright 2009; Albright and Hayes 2003; Bailey and Hahn 2001; Coetzee and Pater 2008; Coleman and Pierrehumbert 1997; Daland et al. 2011; Freyer 2017; Frisch, Pierrehumbert, and Broe 2004; Gorman 2013; Greenberg and Jenkins 1964; Hayes and Wilson 2008; Ohala and Ohala 1978, 1986; Pierrehumbert 2001, 2003; Pizzo 2015; Scholes 1966; Vitevitch and Luce 1998, 1999; Vitevitch et al. 1997; Wilson and Gallagher 2018, amongst others). The general observation made since the earliest literature on the topic (that I am aware of) has been that acceptability judgements form a cline or gradient, *i.e.*, they are not simple binary intuitions. For example, for English speakers, /blik/ » /bnik/ » /bnzk/ (Chomsky and Halle 1968).

In the last few decades, there has been a huge surge of interest in trying to understand such gradient phonotactic judgments, and many such studies now often include sophisticated computational modelling of the human judgments obtained. Of course, it has long been recognised that linguistic performance, including acceptability judgements, is a multifactorial problem, with grammatical knowledge playing only a partial role in it (Chomsky 1965; Du and Durvasula 2025; Schütze 1996; Valian 1982). For example, Kahng and Durvasula (2023) show that perceptual gradient has a sizeable effect on acceptability judgments. In a more stark example, Coleman and Pierrehumbert (1997) observed that <mrupation> was rated quite high by their participants despite the nonword having a completely unobserved sequence [mr] that is generally thought to be ungrammatical as it is a member of a class of other #nasal+consonant sequences that are also unobserved — which they attribute to the presence of a high frequency English (morphemic) sequence <-ation> ending. As Hay, Pierrehumbert, and Beckman (2003) point out, participants often decompose input nonwords into morphemes/words. Given the above, gradient in acceptability judgements is inevitable, and it is likely that a non-trivial amount of it can possibly be explained in terms of non-grammatical (or performance) factors.¹ Consequently, the presence of (sub-

¹For example, one could extend this line of reasoning to explain the trichotomy between /blik/ » /bnik/ » /bnzk/ as resulting from the interaction of a binary phonotactic grammar and morphophonological knowledge — the phonotactic grammar assigns only the first to be grammatical; however, /bnik/ has a rhyme that is identical to the morpheme «-ic». So, following the observation in Coleman and Pierrehumbert (1997), it is

stantial) gradience in acceptability judgements by itself doesn't tell us anything about the underlying grammar without additional work. It requires more careful model comparisons — a fact that has come to the forefront in much recent work, as will be discussed below (Durvasula 2020; Gorman 2013; Kostyszyn and Heinz 2021; Mayer *in press*).

At this point, it is important to clarify what I take to be the main issue when it comes to the phrase *gradience* (or *categoricity*) of the grammar. There are at least two different ways to understand it. The first way to understand the phrase *gradient grammar* is in terms of the EXTENSIONS of grammars having gradient assignments in some form. Specifically, with respect to phonotactics, this view would translate to whether or not there was gradience in the grammaticality of novel wordforms. A second way to understand the word *gradience* is in terms of the INTENSIONAL DESCRIPTIONS within the grammars, and relates to whether or not non-gradient intensional descriptions (or constraints) are possible within a grammar. Again, specifically, with respect to phonotactics, this latter view would translate to asking whether there are any gradient statements/constraints *within* the grammar. While the first way focusses on the output of the grammar, the second way focusses on the structure/-format of the grammar itself. Furthermore, though these two notions of gradience often remain conflated in many discussions of phonotactics or gradient grammars more generally, they are quite different in character and focus. As will be observed later (in the case of the minimal (categorical) phonotactic model presented in this paper), a grammar can be intensionally categorical, while still simultaneously being extensionally gradient. Following Chomsky (1986b, *et seq.*), I take the object of inquiry to be the structure/format of the intensional system itself (namely, the grammar), and not quite the output of the intensional system. Consequently, the appropriate notion of gradience that is of interest in this paper is that of the *intensional* view of *gradience* and whether or not grammars consist of statements/constraints that make reference to gradient information.²

Returning to the topic most relevant to the current paper, what is at stake with modern computational models is to see if the actual observed gradience in acceptability judgements (note, a third notion of gradience — what may be called OBSERVED gradience) can be predicted. These computational modelling attempts can be classified into a few types: (a)

possible to view the improved acceptability of /bnrk/ over /bnzk/ as stemming from the possibility of that false morphological parse. And if the amelioration in acceptability judgement due to the false morphological parse doesn't cancel out the unacceptability from the ungrammaticality perfectly, then one would observe gradience in the judgements.

²A similar dichotomy between intensional and extensional comparisons is present in discussions of *simplicity* and *specificity* in the case of grammar learning/acquisition, as discussed by Durvasula and Liter (2020).

some models presuppose, in contrast to the above discussion, that the observed gradience in judgements *primarily* comes from within the grammar (so they interpret the observed gradience as evidence of intensional gradience) (Albright 2009; Albright and Hayes 2003; Coetzee and Pater 2008; Coleman and Pierrehumbert 1997; Daland et al. 2011; Freyer 2017; Frisch, Pierrehumbert, and Broe 2004; Hayes 2000; Hayes and Wilson 2008; Mayer *in press*; Pierrehumbert 2001, 2003; Pizzo 2015; Shademan 2006; Wilson and Gallagher 2018); (b) alternatively, other researchers have proposed that observed gradience reflects similarity to existing words, perhaps because the judgment task makes use of a lexical search procedure (Bailey and Hahn 2001; Chomsky and Halle 1968; Ohala and Ohala 1978, 1986; Vitevitch and Luce 1998, 1999; Vitevitch et al. 1997); (c) in contrast to the above two modelling strategies, others have shown that computational models with (intensionally) categorical grammars do at least as well (Dai 2025; Durvasula 2020; Gorman 2013; Kostyszyn and Heinz 2021; Swanson, Heinz, and Rawski *accepted*).

Based on the above, the key issues of interest for this paper are to understand what we can learn about the underlying phonotactic knowledge and how it is organised (*i.e.*, what is the nature of the INTENSIONAL DESCRIPTIONS themselves) from such computational models and to understand the source of the sometimes good fit of computational models that include an underlyingly gradient phonotactic grammar. There are a few initial issues that we must address. First, some have argued that there is no phonotactic grammar at all (Reiss 2017), based on the argument that phonotactic acceptability judgements (even if they are systematic) are not in and of themselves evidence of underlying grammatical knowledge, and actually reflect metalinguistic awareness similar to awareness of rhyme or metrics for poetry and parsing. In this case, the question of the categoricity/gradience of such a phonotactic grammar becomes moot. For the purpose of this article, I will mainly presuppose that a phonotactic grammar exists, but I return to this possibility in the Conclusion.

The second issue we need to address is that the proliferation of computational models has ironically, in my opinion, made it difficult to see how one makes progress on the issues, and has sometimes muddled the waters. For example, in very recent work, in responding to the success of categorical models of phonotactics, Mayer (*in press*) has suggested that the distinction between gradient and categorical models is a somewhat superficial dichotomy. This claim is based on the argument that both can be represented or notated using the formalism of *monoids*. However, the presence of a single formalism that can represent both types of grammars doesn't mean that the distinction is a superficial one. What is important are the *properties* of the grammars. One can show the error in the claim in a variety of ways.

For example, any n -dimensional system can be described with an $n+k$ -dimensional system (where $k>0$, and the n dimensions are the same across the comparison).³ This is readily comprehensible by making the additional k -dimensions have a constant value. But, from this, one should not infer that the dimensionality (n vs. $n+k$) is a superficial dichotomy, since the properties of the n -dimensional system and those of the $n+k$ dimensional system are going to be different.⁴ Closer to the issue of computation is the observation that the distinction between finite state machines and Turing machines is not a superficial dichotomy just because the latter can be used to represent the languages generated by both the former and the latter. Note, the properties of the two grammars are different. In contrast, there is no distinction between the properties of context-free grammars vs. push-down automata or finite state languages vs. regular languages, even if the notational systems look different (Lewis and Papadimitriou 1998); therefore those distinctions are superficial or false dichotomies in a real sense. Returning to the question at hand, the distinction between categorical and gradient grammars is indeed a true and important dichotomy given that they have different sets of properties (*e.g.*, they have different classes of possible grammars), and therefore it is important to probe the issue both theoretically and empirically.

The third issue we need to address is that there needs to be a clear basis to assess computational models of phonotactics. A general issue here is that more complex (super-set) models will always fit data better through over-fitting.⁵ This is of course the issue pointed out in footnote 3. We could use a variety of “corrections” (information criteria, prior weighting, *etc.*), but such models, at best, provide a compact descriptions of the corpus/lexicon/training data, without a clear pathway to make inferences about actual human knowledge, which is the real question of interest to a linguist, particularly of the generative ilk.⁶

³This is identical to saying that the R^2 of a statistical model is non-decreasing, *i.e.*, it cannot decrease, but can increase, with the addition of meaningless terms in the model (Cruise 2019). Therefore, R^2 is not a meaningful measure of model fit by itself — without an evaluation of and correction for model complexity, it is not possible to do model comparison.

⁴This is of course what makes *Flatland* (Abbott 1884) so entertaining.

⁵As John von Neumann is (potentially apocryphally) said to have claimed: “With four parameters I can fit an elephant, and with five I can make him wiggle his trunk” (see “Von Neumann’s Elephant” 2025, for discussion).

⁶A reviewer suggested that this claim is a very strong claim, since phonotactic knowledge is usually seen as a compact description, and statistical models themselves are compact descriptions of data. However, such a statistical model can only compress the data in some way, but it doesn’t actually tell us that that is the way humans do it, which is the question we are interested in. There are potentially an unbounded set of data compression techniques; therefore, data compression itself doesn’t tell us anything. For related discussion on the difference between good statistical techniques and comparison to human performance, see a recent

A strategy that has been adopted in the literature to assess computational models is that of comparing model predictions to human judgments. The basic premise is that it is important to identify the computational generalisations potentially learnt by a speaker from a pre-existing corpus, and then test the predictions of the model against human acceptability judgements. I will adopt this general strategy in this paper, as it addresses the question of interest related to what human beings know and naturally ameliorates (but not necessarily, removes) issues related to over-fitting raised above. It is worth highlighting that predictions from models themselves are not meaningful, and different models can indeed have different (numerical) predictions; what is more meaningful (given our current state of knowledge) is the *comparison* to human judgements through some metric of correlation, which focus more on the directions of change in predictions, as will be discussed shortly.

The fourth issue is that we need to be clear about what can be inferred when there is in fact a good fit between computational model predictions and human judgements. It has often been assumed that a good fit of model predictions to linguistic data or to human judgements by themselves show us that the theoretically relevant aspects of the model are justified. However, computational models often come with a host of auxiliary assumptions along with claims about the underlying grammar. Therefore, as a matter of logic, one can't automatically assign blame or credit to the crucial theoretical question as opposed to the auxiliary hypotheses for any theoretical comparison, not just computational modelling (Duhem 1954; Lakatos 1970; Quine 1951). Note, this erroneous argument form must be accepted if one wants to argue that the high performance of a computational model incorporating gradient constraints should be interpreted as evidence that the underlying grammar itself must have gradient knowledge.

With regard to computational models of phonotactics, one could even pose a starker counter-factual question: what if complex gradient models were *better* than the categorical ones in the literature, should we interpret that as evidence for gradient phonotactic knowledge in speakers? In this paper, I will argue that the right answer to this question is “no”. In fact, as I will argue, we need to do more than this to achieve a deeper understanding of underlying phonotactic grammars when comparing computational models. Specifically, I will argue that researchers have to carefully *deconstruct* the computational models to try and truly understand what is leading to the performance. Consequently, simply fitting model predictions against human performance by itself is quite unlikely to further our understanding of these models, and is consequently unlikely to develop well motivated theories

paper on normalisation techniques used in speech perception by Persson, Barreda, and Jaeger (2025).

of phonotactics.

The final issue we need to address is that, as mentioned above, there is a panoply of computational models of phonotactics. Furthermore, as pointed out above, one can't simply compare different computational models that come with a host of other modelling decisions/parameters. In order to make progress, I will use the Hayes & Wilson (Maximum-entropy based) Phonotactic Model as the computational model reflecting a gradient phonotactic grammar (Hayes and Wilson 2008). I chose this particular computational model because it has generally shown good fits to human acceptability judgements. Moreover, the very fact that the model routinely figures in model comparisons against other computational models of phonotactics is a loud (and nearly unanimous) acknowledgement of the impressiveness of the model and the *de facto* baseline that it has now become. Furthermore, the underlying Maximum Entropy grammar (Goldwater and Johnson 2003) has also been used to model a variety of phonological patterns.

There are five main results I will present: (a) the Hayes & Wilson Phonotactic Model model fits do not worsen when the gradient predictions of the model (*i.e.*, extensional gradience) are effectively binarised (Section 3.2), (b) the gradience in the predictions of the Hayes & Wilson Phonotactic Model doesn't stem from the constraint weights, as a modified model with no constraint weights is at least as well correlated with human acceptability judgements (Section 3.3), (c) the gradience in the predictions of the Hayes & Wilson Phonotactic Model doesn't stem from the lexical (segment-based) type frequency in the training data, as a model trained on training data with no (segment-based) frequency structure/differentiation between the consonant sequences are at least as well correlated with human acceptability judgements (Section 3.4), (d) nearly degenerate Hayes & Wilson Phonotactic Models (one of them with just six constraints) and no differentiation in constraint weights are at least as well correlated with human acceptability judgements (Section 3.5) (e) in addition, in Section 4, I will describe a new model (the **minimal (categorical) phonotactic model**) and then show that its performance, when correlated against human judgements, is virtually identical to that of all the other models despite being minimally specified, and *intensionally* categorical. This final result suggests that even if one is committed to the view that some gradience in acceptability judgements is due to the grammar, it doesn't mean that the grammar itself is gradient, *i.e.*, extensional gradience doesn't entail intensional gradience.

2 Modelling decisions

2.1 Training data

In order to keep as many modelling choices constant as possible and thereby maintain the *all things kept equal* clause of the reasoning, the core training data I will use, following Hayes and Wilson (2008), is a modified version of list of word-initial consonant sequences in the CMU dictionary (Weide 1994). As they note, the list excludes any consonant sequences that they judged to be exotic.⁷ The list of consonant sequences and their type frequencies are listed in Table 1.

Onset	Type Freq.	Onset	Type Freq.	Onset	Type Freq.	Onset	Type Freq.
k	2,764	v	615	j	268	z	83
ɹ	2,752	g	537	fɹ	254	sm	82
d	2,526	dʒ	524	pl	238	θɹ	73
s	2,215	st	521	bl	213	skw	69
m	1,965	tɹ	515	sl	213	tw	55
p	1,881	kɹ	387	dɹ	211	spɹ	51
b	1,544	ʃ	379	kw	201	ʃɹ	40
l	1,225	gɹ	331	stɹ	183	spl	27
f	1,222	tʃ	329	θ	173	ð	19
h	1,153	bɹ	319	sw	153	dw	17
t	1,146	sp	313	gl	131	gw	11
pɹ	1,046	fl	290	hw	111	θw	4
w	780	kl	285	sn	109	skl	1
n	716	sk	278	skɹ	93		

Table 1: Onsets in broad IPA transcriptions and type frequency counts used by Hayes and Wilson (2008)

2.2 Test data

I will focus on acceptability judgements of English words where the computational modelling involves the word-initial consonant sequences. One might wonder if it is more appropriate to compare *whole-word* models. I personally am skeptical this is appropriate as whole-

⁷I will return to elaborate on this issue in the Conclusion.

words consist of much more representational structure than just segment sequences (if we accept the research in phonology since the 1950s, which I am wont to do). Therefore, unless we are able to model the word in a phonologically sensitive way, there is a distinct danger that conflating/ignoring important representations can lead to spurious fits. In fact, if anything, even the category of word-initial consonant sequences might be conflating important representational differences that phonologists have argued for (Goad 2012; Kahn 1976; Scheer 2004); however, for the purposes of the paper, it is assumed based on the most standardly assumed syllable structure (Blevins 1995), without further justification, that the relevant word-initial consonant sequences form a natural class of a reasonably simple representational unit, namely, *onsets*.⁸ Furthermore, there appear to be many other effects on judgements apart from the grammar. For example, as mentioned above, Kahng and Durvasula (2023) found evidence of perceptual gradience affecting judgements. As a second example, Coleman and Pierrehumbert (1997) observed that <mrupation> was rated quite high by their participants despite the nonword having a completely unobserved sequence [mr], which they attribute to the presence of a high frequency English (morphemic) sequence <-ation> ending. As pointed out earlier, no current grammatical model predicts this. In fact, it is not even currently clear how one formally models such ameliorating effects to otherwise unacceptable nonword sequences. Therefore, I think it is somewhat inappropriate to test the whole-word predictions of computational models.

Beyond the general issues raised above, the topic of acceptability judgements of English words where the crucial variable is the word-initial consonant sequences forms an extremely focussed problem that does not conflate effects of syllable structure, onset-vowel interactions, *etc.* Furthermore, this was one of the main foci of Hayes and Wilson (2008), and equally importantly, at this point, there are multiple acceptability judgement studies related to this topic. Specifically, I will use data from (a) Scholes (1966), (b) Albright (2007), (c) Daland et al. (2011). As discussed below, the three different experiments had a variety of different stimuli, experiment designs, and even prompts. A consistent result across all the experiments therefore allows us to, to some extent, sidestep the problem that the inference might otherwise be affected by important considerations related to experiment design, namely the nature of the task (Armstrong, Gleitman, and Gleitman 1983) or the implications of the exact prompts (Firestone and Scholl 2016; Schütze 2005).

The Scholes (1966) data is based on 35 native English speaking seventh graders who

⁸Incidentally, Albright (2009) observed that the models performed better when trained on onset-only data.

responded Yes-No to 66 monosyllabic nonwords varying in consonant sequences, where the rhymes were all bland. The prompt used was “Suppose these are foreign words which English wishes to borrow; which ones will be admitted in their present form and which ones will be changed?” The stimuli were presented auditorily to all students in the classroom at the same time.

The Albright (2007) data is based on native English speaking participants who judged 30 monosyllabic nonwords varying in consonant sequences starting with p-, b- on a 7-step Likert scale, where 1 indicated “Completely impossible as an English word” and 7 indicated “Would make a fine English word”. The rhymes were controlled for neighbourhood density and bigram probability. The task also included 170 filler items (nonce+real). The stimuli were presented auditorily using PsyScope (Cohen et al. 1993) in an in-lab experiment. As an interesting detail, participants were asked to repeat each nonword aloud before rating it — the participant pronunciations were used to exclude any token (for that participant) if the participant didn’t achieve the intended target.

The Daland et al. (2011) data is based on 48 native English speaking participants who judged 96 stress-initial CCVCVC non-words with 48 unique word-initial consonant sequences on a 7-step Likert scale, where 1 indicated “unlikely” and 7 indicated “likely”. The participant were told to “rate them based on how likely it was that the words could become new words of English in the 21st century”. The tail -VCVC sequences were not balanced across stimuli, and consisted of 6 different sequences [atɪf, ɪbɪd, ʌspɪ, ɛpɪd, ɪɡɪf, ɛzɪɡ] depending on the word-initial consonant sequence. The participants were recruited through Amazon Mechanical Turk, and therefore the experiment was an online (out of lab) experiment. Finally, the stimuli were presented to the participants orthographically.

As mentioned above, following Hayes and Wilson (2008), I modelled only the word-initial consonants for acceptability judgements from each of the three experiments. However, each data set had multiple words with the same word-initial consonant sequences, but with different rhymes. To prevent double counting of the consonant sequences and thereby potentially bias the correlations presented, the participant judgment values for each unique consonant pair were averaged.⁹ These averaged values are presented in Figures 1-3. As can be seen across all three experiments, there is a cline of judgements. And it is the case in all three experiments that the observed word-initial sequences are rated generally higher than the unobserved ones. In the figures in this paper, *grossStatus* indicates whether the

⁹It is worth noting that the ratings in all three experiments for stimuli with identical word-initial consonant sequences were very similar. Therefore, the concern was that any measure of fit would be inflated due to the presence of the repetition.

sequence is observed (Observed) or not (Unobserved) as a word-initial consonant sequence in the training data mentioned above.

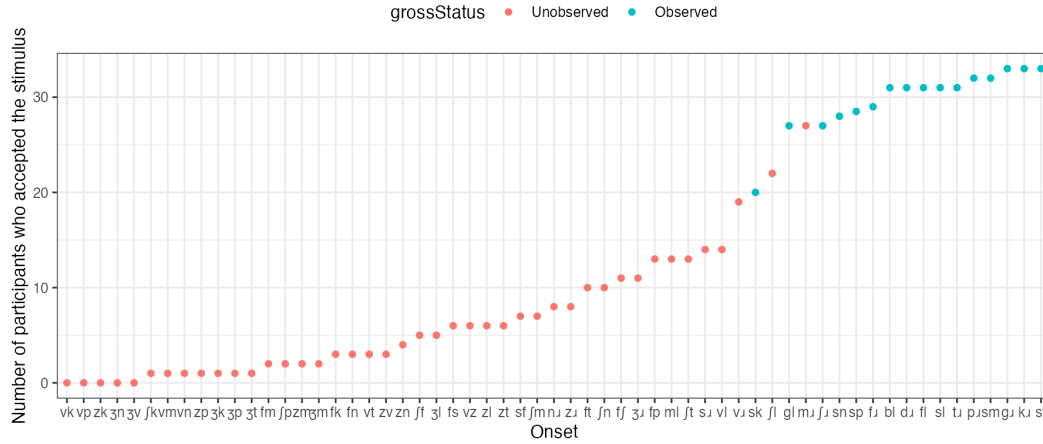


Figure 1: Overall judgements in Scholes (1966). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

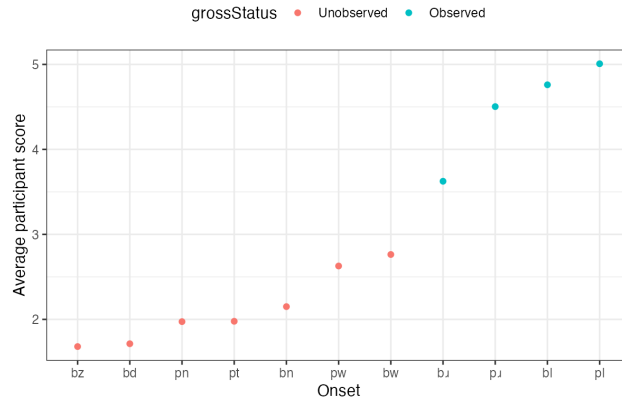


Figure 2: Overall judgements in Albright (2007). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

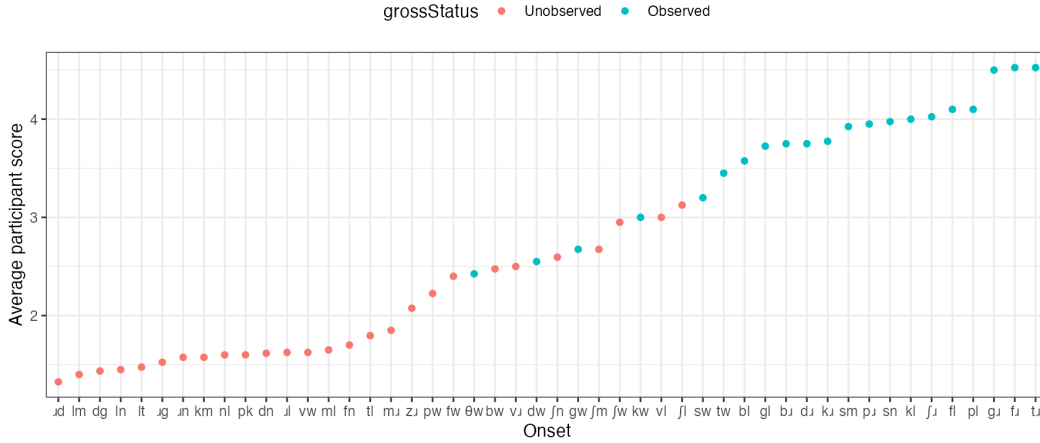


Figure 3: Overall judgements in Daland et al. (2011). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

2.3 General remarks about the presentation of results

In this paper, I used the `tidyverse` suite of packages (Wickham 2017) for data analysis and plotting. The statistical analysis was done in base R (R Core Team 2021). Finally, the data and grammars (constraint sets) were converted into $\text{\LaTeX}$ code using the R-package `stargazer` (Hlavac 2018).

Throughout, I will use correlation measures as measures of fit to human judgements because they side-step the issue of the relevant scale of the predicted/observed judgements.¹⁰ Note, in the case of Pearson’s correlation, this would mean that differences in values between measures that are linear transforms of each other are not relevant differences for us. As a guiding intuition for this choice, it is worth thinking about the measurement of temperature in terms of Fahrenheit and Celsius, which are linear transforms of each other — it is not an appropriate inference that models that predict temperature in Fahrenheit are different in terms of fit to actual temperatures from those that predict temperature in Celsius. Different measures of correlation get at this intuition, though they make different assumptions about the nature of the relationship. Specifically, I will use both Pearson’s correlation co-efficient (r) and Kendall’s τ_b as the measure of fits. I include Pearson’s r largely because using non-parametric measures militates against a crucial distinction made in the Hayes & Wilson Phonotactic Model. Hayes & Wilson Phonotactic Model distinguishes

¹⁰I will elaborate on this issue more when discussing some of the results later.

between a *harmony* score (a weighted score of constraint violations) and the MaxEnt score or negative exponentiated score, the latter of which is used in lieu of a probability value associated with the test sequence.¹¹ Using a non-parametric correlation measure will not allow us to distinguish between harmony scores and the negative exponentiated scores, since the negative exponentiation simply reverses the rank-order of the harmony scores.¹² Therefore, to maintain a core theoretical distinction of the Hayes & Wilson Phonotactic Model, the Pearson’s r was used as a measure of model fit. However, Pearson’s r comes with its own share of technical issues in that it assumes a linear relationship between the variables, and assumes that the errors are Normally distributed. Though, as will be seen in the figures presented, the relationships are reasonably linear,¹³ using the Pearson’s r is not appropriate if the errors are not normally distributed; therefore, I have included the non-parametric Kendall’s τ_b too.

Both Pearson’s r and Kendall’s τ_b , like a mean, are just descriptive statistics which don’t allow for inferences on their own without associated error probabilities. So, instead of just looking at which model has the highest correlation coefficients as has been the case in (all) prior literature, I will consider the 95% confidence intervals for the correlation coefficient in order to see if a specific fit is statistically different from another. However, one can’t just check if two confidence intervals overlap as that procedure has been known to be conservative (favouring the null hypothesis). For example, when the bottom confidence interval of one estimate coincides with the top confidence interval of another estimate, then the difference is significant at a much lower alpha value than alpha=0.05 (Cumming and Finch 2005). A lower than alpha=0.05 is not unreasonable when trying to establish a

¹¹I ask for the reader’s forbearance in understanding this distinction for now, and will return to it when the model is elaborated in Section 3.1.

¹²Just to belabour the point with a concrete example. Just imagine a set of values, and another set of the same values that are negative exponentiated. One can easily run the following code in R to see that the Pearson r values are different for the original and negative exponentiated values, while the Kendall and Spearman coefficients are identical for the two types of values:

```
X=rnorm(100); Y = X+rnorm(100); X_negExp=exp(-X); Y_negExp=exp(-Y);
```

```
#Comparing correlations of original values and negative exponentiated values.
```

```
cor(X,Y, method="pearson")-cor(X_negExp,Y_negExp, method="pearson") # ≠0
```

```
cor(X,Y, method="kendall")-cor(X_negExp,Y_negExp, method="kendall") # =0
```

```
cor(X,Y, method="spearman")-cor(X_negExp,Y_negExp, method="spearman") # =0
```

¹³The relationships are reasonable linear for harmony scores, and as will be argued are effectively binarised when negative exponentiated. So in both cases, the linearity assumption is reasonable.

new result, particularly for the Hayes and Wilson (2008) model since the actual model fit estimates vary with different runs on the same training and testing data (as is the case with the Hayes & Wilson Phonotactic Model model). However, here I will stick to $\alpha=0.05$. As argued by Cumming and Finch (2005, p. 176), “For a comparison of two independent means, $p \leq 0.05$ when the overlap of the 95% CIs is no more than about half the average margin of error, that is, when proportion overlap is about 0.50 or less.... when both sample sizes are at least 10, and the margins of error do not differ by more than a factor of 2”. The relevant samples sizes are all above 10, and the margins of errors do not differ by a factor of 2. Therefore, this is ultimately the decision criterion in this paper. Furthermore, as a quick heuristic, one can check if the 95% confidence intervals of one estimate traps the other estimate; if so, then the two estimates are definitely not significant at $\alpha=0.05$. In case, where this is not the case, one can use the more precise test of proportional overlap of the confidence intervals.

Furthermore, I will present all the figures as the model’s predictions against human performance with the model predictions on the x-axis (therefore, treating it as the independent variable). While this does not make a difference to the measure of fits in this paper, it does affect any linear modelling of the relationship. As pointed out by Piñeiro et al. (2008), it is mathematically appropriate to model the observed values in terms of predicted values (so, the latter is on the x-axis in figures), as opposed to the other way around, since the latter leads to spurious regressor effects. Furthermore, the same modelling and visualisation choice was made by Hayes and Wilson (2008).

Finally, a reviewer argues that the predicted and observed judgements need to be “quite parallel” for there to be evidence that a model is on the right track. First, in practically every previous study on a comparison between a phonotactic model and human judgements, a measure of correlation was used (Dai 2025; Gorman 2013; Hayes and Wilson 2008; Kostyszyn and Heinz 2021; Mayer *in press*); so, I believe that the use of a correlation measure here continues the tradition and is appropriate; consequently, X-Y plots presenting such relationships are maintained in the article. Second, and more importantly, an expectation that the predicted and observed judgements be “quite parallel” means of course that the grammatical model predicts the judgements, except for some constant difference (and modulo some noise) — but this is true only if the predicted and observed scores are on the same scale, and not otherwise. Furthermore, if one were to push the parallel expectation reasoning to its logical conclusion, it is not clear why one would even want a constant difference and not overlapping values instead? There is a supposition here that a constant difference

is somehow OK, but it is important to point out that such an assumption is not justified in any formal or theoretical way. As mentioned above and elaborated on later, the more appropriate expectation is that two sets of values are similar once differences in scales are removed (as such differences can easily stem from non-crucial differences in what numerical values are associated to being a violation versus not). Third, perhaps, one way to understand the “quite parallel” expectation, is as a claim that non-grammatical factors add (at best) a constant value to the judgement of any stimulus. Such an underlying claim is wrong based on the fact that we already have good reason to believe that extragrammatical effects do not affect judgements uniformly across all stimuli — this has already been discussed in the article.

3 Understanding the original Hayes & Wilson Phonotactic Model performance results

3.1 Background on the Hayes & Wilson Phonotactic Model

The primary intuitions that the Hayes & Wilson Phonotactic Model is based on are as follows. Using the actual observed lexicon, the model extracts featural ngram generalisations or constraints and figures out weights to assign to constraints (using gradient descent) — this weights are generally thought to be related to the frequency of occurrence of the ngram in the lexicon (see Hayes and Wilson 2008, for details). The constraints and the corresponding weights are used to get a *harmony score*, which is simply the sum of the products of the constraint violations and the corresponding constraint weights, as in (1).

1. The harmony score of a candidate

$$h(x) = \sum_{n=1}^N w_n C_n(x)$$

N = number of constraints

$C_n(x)$ = violation score associated with a constraint_n

w_n = weight associated with constraint_n.

Finally, to produce a “judgement” for a nonsense word, the harmony score is then converted into a probability using the formula in (2).

$$2. P(\text{candidate}) = \frac{\exp(-\text{harmony score})}{\sum_{n=1}^{n=\text{all possible words}} \exp(-\text{harmony score}_n)}$$

However, to calculate the denominator in the formula in (2), we need the relevant values of all the possible words in a language — which while theoretically an infinite set, has to be finite for a probability estimation.¹⁴ In practice, however, the denominator is ignored, as it is a constant across all the stimuli, and behaves as a linear scaling factor, so the sum of all scores adds up to one to satisfy one of the axioms of probability. Instead, just the numerator is used as a prediction of the model — this is called the MaxEnt score. I will call the numerator term the *negative exponentiated* score throughout.

Though not often discussed, there are four substantive/theoretical claims in the model, as listed in (3). Out of these, Claim 3 is perhaps the most novel observation that will become important when we look at the model fits.

3. Substantive claims in the Hayes & Wilson Phonotactic Model

Claim 1: The grammar has gradient knowledge within it (*i.e.*, constraints are weighted).

Claim 2: The effect of the constraints is cumulative (additive in the case the harmony score, and multiplicative in the case of the negative exponentiated scores).

Claim 3: The predicted harmony scores are effectively binarised (a claim that will become clearer later in the paper).

Claim 4: The probability of a particular form is in comparison to all other possible word-forms.

Beyond the above substantive claims, the model also includes many other (parametric) decisions: (a) feature set used for training, (b) the maximum number of constraints to discover; (c) the window size of the constraints (1,2,...); (d) the Observed/Expected (O/E) ratio as a measure of constraint effectiveness that is used as part of the constraint induction; (e) allowing negation or complement classes (*e.g.*, from the manual that accompanies the software “Example: *[^-voice,+ant,+strid][+nasal], as applied to English, means ‘assess a violation for any nasal preceded by a segment other than [s].’”); (f) the possibility of tier projections.

For the purposes of this paper, I made the following (parametric) decisions. First, I used a fully specified feature set (based on the original contrastive feature set used by Hayes and

¹⁴This step is of course theoretically suspect, and definitely casts a doubt on such models, because it raises the question of how a human identifies such probabilities then. Furthermore, this is part of a larger general theoretical problem with computational models of phonotactics that see word acceptability as a manifestation of a probability value. I touch on this issue briefly in the Conclusion.

Wilson (2008)) where a negative or 0 feature value was represented as the negative value. I did this as claims of underspecification, which I am deeply sympathetic to, themselves should be tested separately. Second, the models have a maximum of 20 constraints (note, Hayes and Wilson (2008) obtained 23). Third, the maximum window size was 2. Fourth, the O/E criterion was 0.35. I used this value both because it was less than 0.5 (which seemed like a reasonable minimum criterion to me) and because 0.35 is the default number in the manual that accompanies the software (Capodiceci, Hayes, and Wilson 2009); furthermore, there is no clear independent reason to choose another O/E criterion. Fifth, I did not include the possibility of complement classes, as that is quite atypical given the work on constraint definitions in either Optimality Theory or Rules-and-Constraints theories. Finally, I did not include any tier projections as Hayes and Wilson (2008) didn't include them in their modelling of the relevant English onset phonotactic facts. It might strike the reader that some of these decisions are either arbitrary or unfair. But, it is important to note that the objective here is primarily to control factors as much as possible when comparing different models; and given the number of adjustable (free) parameters, it is not feasible to try out every combination of possible parametric choices. Having said that, in Section 3.6, I return to this issue and compare the basic parameterisation with a different set of parameters (a maximum of 100 constraints, window size=2, O/E = {0.08,0.8}, including complement classes, and using the original contrastive feature set that Hayes and Wilson (2008) used) and show that the (parametric) choices made for the purposes of this paper are quite reasonable and don't worsen the predictions compared to other possibilities.

3.2 Results

In this section, I will present the performance of the Hayes & Wilson Phonotactic Model against the three sets of experimental results. The model comparisons will be presented as figures, as all the relevant information is contained in them. For each of the three experimental results, I will present the model's predictions in terms of negative harmony scores (*-h scores* in the figures), and as negative exponentiated scores (*exp(-h scores)* in the figures).¹⁵ I present negative harmony scores on the recommendation of a reviewer who thought it more intuitive that a higher value represented a more grammatical sequence. Furthermore, the two measures of fit (Pearson's r and Kendall's τ_b) along with the corresponding 95% confidence intervals are presented at the top of the facet corresponding to the relevant score.

¹⁵Note, the full set of all predicted scores for all the models presented in this paper is available in Appendix A.

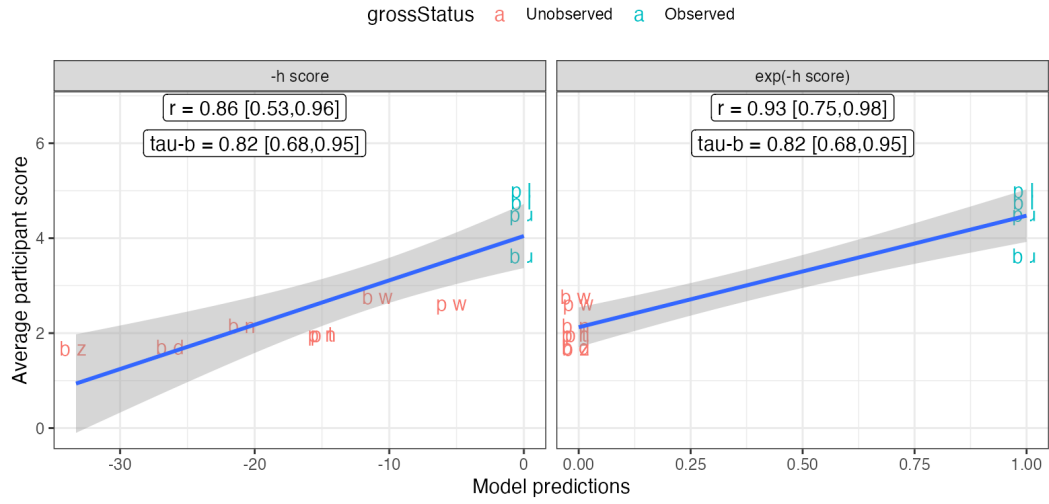


Figure 5: Comparison of the Hayes & Wilson Phonotactic Model predictions to human judgements for the Albright (2007) data. Numbers in square brackets refer to the 95% confidence intervals.

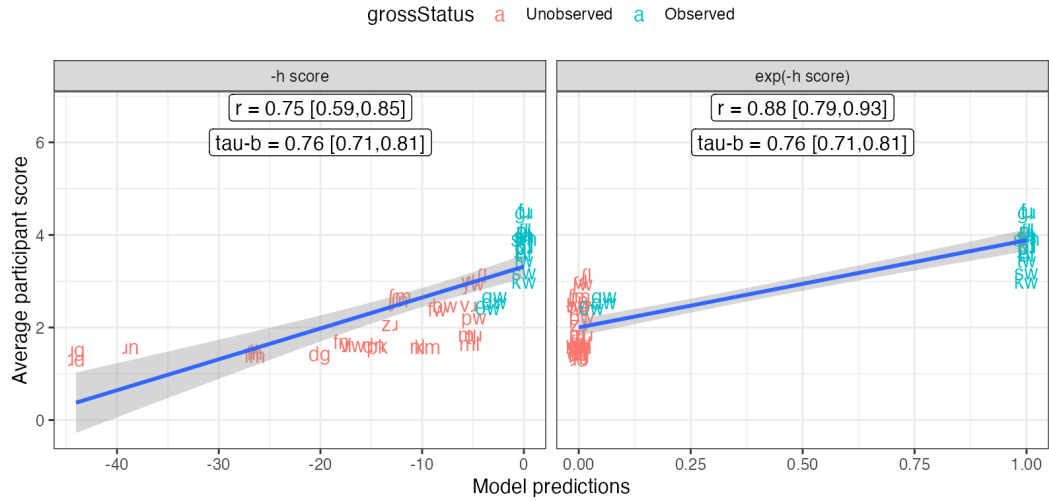


Figure 6: Comparison of the Hayes & Wilson Phonotactic Model predictions to human judgements for the Daland et al. (2011) data. Numbers in square brackets refer to the 95% confidence intervals.

Note, the right panel in Figure 4 looks different from a similar plot in Hayes and Wilson (2008) because they applied an additional free parameter (temperature parameter) to improve the fit that was not used here for reasons mentioned above. Furthermore, as pointed out above, Kendall’s τ_b will have the same values for the -h scores and the negative exponentiated scores. However, with respect to Pearson’s r , the figures make clear that the fits are generally improving with negative exponentiation when the transform results in *less* gradient in the predictions; essentially, the transform is binarising the predictions. However, it is important to note that none of these differences are significant, when we apply Cumming and Finch’s (2005) criterion that the CIs overlap no more than about half the average margin of error. So, while the raw values of fits do improve similar to what Hayes and Wilson (2008) observed, the improvement is not statistically significant. One could perhaps say that the consistency of the shift (increases in the case of the negative exponentiated values) is some indication of an improve fit; however, even here, since there are only 3 comparisons, it should not be taken as dispositive.

The more important fact for our concerns is the that the effective binarisation of the model predictions (observed in the negative exponentiated scores) still results in an impressive fit: this is our first clue that the gradient predicted by the model is not as impressive as what might be thought based on just looking at the model fits. It therefore raises awareness of the need for much more careful model inspection to understand why the model is successful.

3.3 Model check 1: What if the Hayes & Wilson Phonotactic Model had no gradient constraint weights?

The previous sub-section observed that even when the gradient in predictions decreases (perhaps, even removed) in the predictions of the grammar, there seems to be an impressive fit to human judgements (in fact, numerical better than when the gradient is present). However, the underlying grammar still possessed gradient in the form of constraint weights. So, one could have argued that the observation in the previous sub-section doesn’t directly speak to the gradient/categorical nature of phonotactic constraints themselves. In this sub-section, we address this concern by simply removing the gradient in the constraint weights (which is the same mathematically as making the constraint weights all the same, and equal to 1 in the original model).¹⁶ So, effectively, the Hayes & Wilson Phonotactic Model

¹⁶To maintain a categorical constraint, what really matters is that each constraint results in the same two constant values {m,n} across-the-board corresponding to violation vs. not. For now, I assume that the

now reduces to a modified model where one just sums up the constraint violations for the same grammar (without using the constraint weights). Therefore, the harmony score of a candidate is now modified as in (4), and the subsequent negative exponentiation will be done as before. I will call this modified model the *No-weights* Hayes & Wilson Phonotatic Model.¹⁷

4. A modified No-weights Hayes & Wilson Phonotatic Model

$$\sum_{n=1}^N C_n(x)$$

In Figure 7, the harmony scores of the No-weights Hayes & Wilson Phonotatic Model are compared against the original harmony scores. As can be seen in the figure, despite the constraint weights differing appreciably, the model predictions of the two models are highly correlated.

values are $\{1,0\}$, but I will return to this issue later in the article. See footnote 20 for related discussion.

¹⁷Note, the learner still relies on gradient information about the constraints to learn the constraints. As will be discussed in the Conclusion, the question of how a grammar is learnt should be kept separate from what the grammar is. Here, I am testing the latter question, to the extent possible for the Hayes & Wilson Phonotatic Model model.

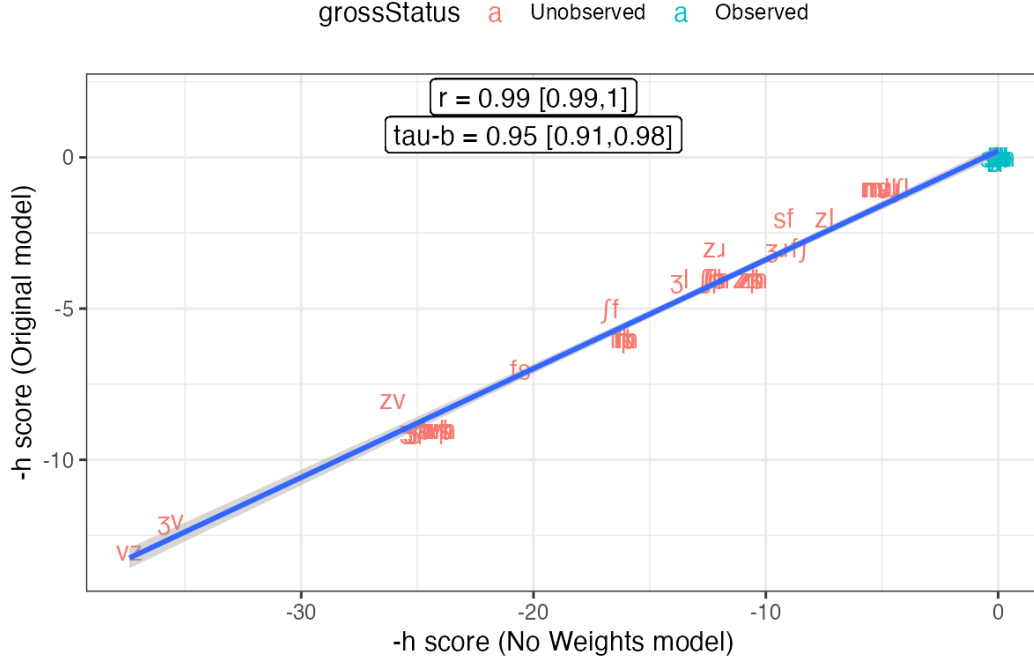


Figure 7: Comparing the predictions of the standard Hayes & Wilson Phonotatic Model with a model with No-weights Hayes & Wilson Phonotatic Model for the Scholes (1966) data. Numbers in square brackets refer to the 95% confidence intervals.

Table 2 has model fits for each of the two models for human judgements for each of the three experimental results studied in this paper. Irrespective of the fit measure used, the No-weights Hayes & Wilson Phonotatic Model has generally *higher* fits, but none of them are statistically significantly higher than that for the original Hayes & Wilson Phonotatic Model, using the Cumming and Finch’s (2005) diagnostic.

	Pearson’s r		Kendall’s τ_b	
	Original	No-weights model	Original	No-weights model
Scholes (1966)	0.88	0.94 [0.90,0.96]	0.71	0.75 [0.66,0.84]
Albright (2007)	0.93	0.94 [0.79,0.99]	0.82	0.82 [0.68,0.95]
Daland et al. (2011)	0.88	0.90 [0.83,0.95]	0.76	0.76 [0.70,0.82]

Table 2: Correlations of the negative exponentiated scores with judgement data in all three experiments. Numbers in square brackets refer to the 95% confidence intervals.

3.4 Model check 2: Where do the weights come from and do we need them?

The previous sub-sections showed two things: (a) the fit of the Hayes & Wilson Phonotactic Model was pretty good against human performance; however, the fit was no worse when the gradient in the *predictions* was *reduced* (or removed even), thereby casting some initial doubt that gradient in the model predictions is truly useful; (b) the gradient in the model constraint weights itself is not useful as a No-weights Hayes & Wilson Phonotactic Model does at least as well as the original Hayes & Wilson Phonotactic Model.

The two results so far suggest that gradient in the predictions and the gradient in the grammar itself don't improve fits — in fact, if anything, they seem to worsen fits (at least, numerically, though not statistically significantly). However, there is still the issue of there being gradient in the training data, in the form of type frequency information about the different word-initial consonant sequences. In this sub-section, we probe this latter issue by attempting to remove such frequency information in the training data.

To do this, we first have to figure out what the right frequency-free training data should look like. Table 3 provides a couple of possibilities that should be entertained. In general, we can't directly compare a model trained on a lexicon with type frequency information with a model trained on a lexicon with unique types (see the first and second columns of Table 3, respectively), because that would be an apples-to-oranges comparison, in that there is a massive difference in the training data size. In the example, the training data size for the unique consonant sequence count would be just 3 compared to the 30 for the type frequency count. Instead there's an argument to be made that we need to compare a model trained on a lexicon with equal frequency weighting, as in the last column of the same table, which has the same total size. The main intuition is that we need to deconfound training data size and frequency structure in the training data. It is only the equal type frequency data set that erases frequency structure (at the segmental level) while retaining the training data size. The same point is made in Sarver (2020). In what follows, I will call the unique consonant sequence count case the *1-frequency* training data; and I will call the latter type the *equal frequency* training data. Note, even these modified corpora have some frequency structure at the featural level (albeit, not at the segmental level), because some features and feature combinations are likely more represented than others in a set of segments or segment sequences. So one could in theory approach the problem differently by re-envisioning the frequency structure over features. I implement the lexicon as described in order to be as faithful as possible to how Hayes and Wilson (2008) approach it in training

their model.

Consonant sequence	Actual type freq.	1-frequency	Equal type frequency
k	15	1	10
r	10	1	10
kw	5	1	10
Total	30	3	30

Table 3: A potential training set of unique word-initial consonant sequences

The constraints and corresponding weights for each of the three models are presented in Table 4. The table also provides the constraint weights from a repeated modelling of the Hayes & Wilson Phonotatic Model with exactly the same parameters as the modelling in Section 3. As Hayes and Wilson (2008) note, the constraint weights will differ between different training sessions, even if all the parameters are kept the same, so it is instructive to see model fits and the constraints extracted under repetition to better understand the differences to other types of models.

The first thing to notice about the different models is that the constraint sets are not identical. There is quite some similarity between the original Hayes & Wilson Phonotatic Model modelling (*Original* in the table) from Section 3.1 and the repeated modelling using the same parameters and training data (*Original (rep)* in the table). However, there is some variation in the set of constraints and the constraint weights.

When one compares the original model to the model trained on the equal frequency training data, the constraint sets and constraint weights are more different. Finally, the model trained on the 1-frequency training data is rather different from the original model — it includes just 6 constraints!

Constraint	Original	Original (rep)	EqFreq	1-freq.
*[+nasal,+dorsal]	5.594	5.594	5.370	
*[+sonorant][-word_boundary]	5.257	5.322	6.575	
*[-labial,-strident,-dorsal][+consonantal]	5.148	5.128	5.783	
*[-spread_gl,-anterior,-dorsal][-distributed,-lateral]	4.848	4.927	6.125	1.931
*[+continuant,+voice][-word_boundary]	4.774	2.824	3.725	2.599
*[+continuant,-labiodental,-distributed][+continuant,-round,-lateral]	4.765	5.078	5.413	
*[-continuant,+delayed_release][-word_boundary]	4.735	4.229	4.830	
*[-word_boundary][+delayed_release]	4.469	4.538	5.082	2.068
*[+distributed][+consonantal]	4.177	4.177	4.800	
*[-word_boundary][-nasal,+voice,-round,-distributed,-lateral]	3.577		4.400	
*[+continuant,-spread_gl,-strident][-distributed,-lateral]	3.563	3.569		
*[+voice][-distributed,-lateral]	2.852	2.854		
*[-sonorant,+continuant,+voice,-labial]	2.696			
*[-word_boundary][-sonorant,+voice]	2.580	5.332	3.086	
*[-strident][-round,-distributed,-lateral]	2.361	4.579	2.924	
*[-strident][-approximant]	2.111	1.718	2.425	2.038
*[-sonorant,+continuant,+voice,+distributed]	2.071	4.049		
*[-anterior][-approximant]	1.935	1.705	2.216	1.791
*[-anterior][-round,-distributed,-lateral]	1.277	1.454	1.455	
*[+voice][-approximant]	0.353	0.445	0.755	
*[+continuant,+voice,+strident]		2.688		
*[+delayed_release,+voice][-word_boundary]		2.639	3.296	
*[+continuant,+voice,-anterior,+strident]			4.789	
*[+voice,+coronal][+consonantal]			1.134	
*[+voice][-round,-distributed,-lateral]			0.784	
*[-labial,-strident,-dorsal][-round,-distributed]				1.838

Table 4: Comparing the constraint weights when the Hayes & Wilson Phonotactic Model is trained with actual type frequency data vs. when it is trained with different type of frequency-free training data. A blank cell indicated the model didn't include that constraint.

In Figure 8, the harmony scores of the models with different types of modified training data are compared against the original Hayes & Wilson Phonotatic Model harmony scores. Despite the fact that the actual grammars and the weights in them are quite different, quite surprisingly, the correlations between the predicted scores are quite good. In fact, the correlations in predicted harmony scores are similar to that between the original scores and the scores from the repeated modelling [$r = 0.99$].

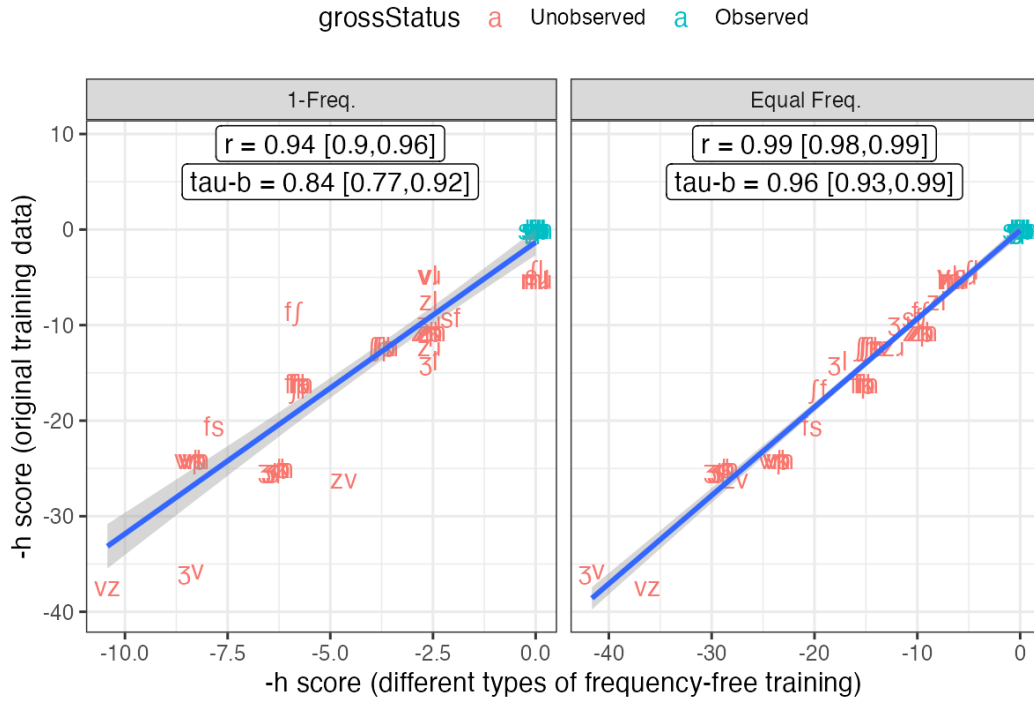


Figure 8: Comparing the predictions when the Hayes & Wilson Phonotatic Model is trained with actual type frequency data vs. when it is trained with different types of frequency-free training data. Numbers in square brackets refer to the 95% confidence intervals.

Table 5 has model fits for each of the modified training data cases and the original model when compared to human judgements from all three experiments. As can be seen, the frequency structure in the training data made little difference to the model fits. In almost every case, the CI's trap the estimates for the original model. The 1-frequency case for the Daland et al. (2011) is the one case where the model fits are significantly lower than the original model. The results generally suggest that (segment-based) frequency structure

is not necessary in the training data to achieve equivalent fits to human judgements.

Correlation type	Data	Original	Equal type frequency	1-frequency
Pearson’s r	Scholes (1966)	0.88	0.88 [0.81,0.93]	0.88 [0.81,0.93]
	Albright (2009)	0.93	0.93 [0.75,0.98]	0.95 [0.82,0.99]
	Daland et al. (2011)	0.88	0.84 [0.72,0.91]	0.68 [0.50,0.81]
Kendall’s τ_b	Scholes (1966)	0.71	0.69 [0.59,0.80]	0.66 [0.55,0.76]
	Albright (2009)	0.82	0.85 [0.70,0.99]	0.87 [0.80,0.95]
	Daland et al. (2011)	0.76	0.72 [0.65,0.80]	0.62 [0.50, 0.74]

Table 5: Correlations of the negative exponentiated scores of different types of modified Hayes & Wilson Phonotatic Models with Scholes’s (1966) data. Numbers in square brackets refer to the 95% confidence intervals. Red text indicates a significant difference.

3.5 Model check 3: Combining 1-frequency training with a No-weights model

As a final model check, in this sub-section, I present the results of a No-weights model that was trained on both the equal frequency and 1-frequency training data. Essentially, we are looking at the model predictions of a model trained on either the equal frequency or 1-frequency training data, namely, the last two columns of Table 4, where the constraint weights are ignored (so, effectively, equal to 1). Therefore, these are extreme versions of the Hayes & Wilson Phonotatic Model where there is no segment-based frequency information in the training data, and there are no model constraint weights either.

As an example, as can be seen in Figure 9, the models’ harmony scores correlate quite well with those of the original Hayes & Wilson Phonotatic Model for the Scholes (1966) stimuli.

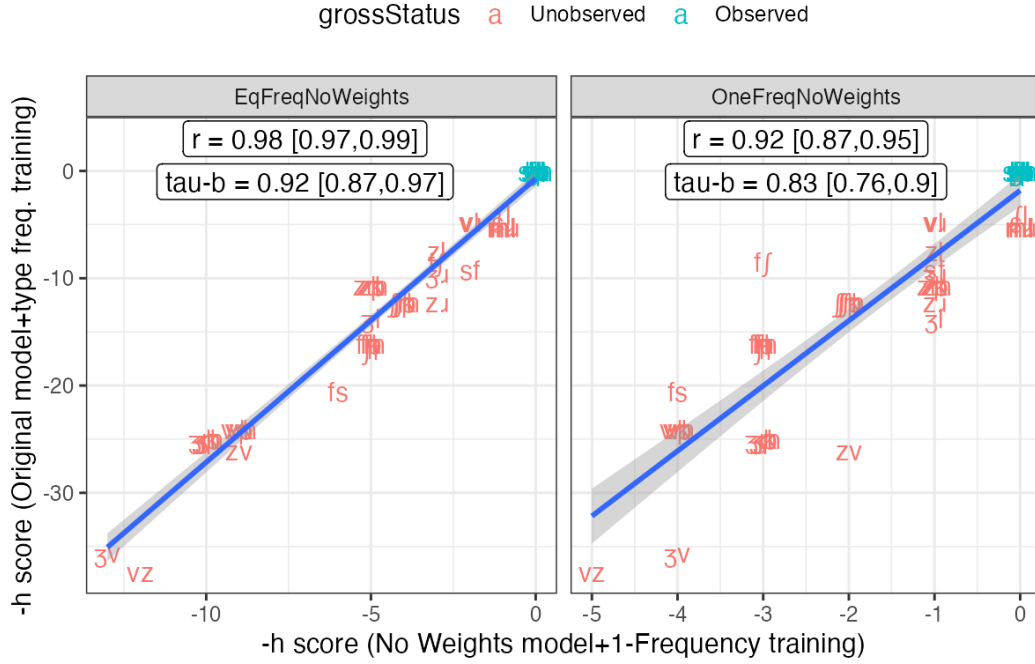


Figure 9: Comparing the predictions when the Hayes & Wilson Phonotactic Model is trained with actual type frequency data vs. a No-weights model trained on both the Equal frequency and 1-frequency training data for the Scholes (1966) stimuli. Numbers in square brackets refer to the 95% confidence intervals.

Furthermore, when we look at the model fits against human judgements from all three experiments (Table 6), in almost every case the CIs trap the estimate for the original model. As in the previous comparison, for the Daland et al. (2011) results, the fit is statistically significantly worse than the original model for the 1-frequency model (in this case, with no-weights). The results do generally suggest that removing (segmental) frequency structure and the weights from the constraints in the grammar don't worsen the fit to human judgements.

Correlation type	Data	Original	Eq. freq. & No-weights	1-freq. & No-weights
Pearson’s r	Scholes (1966)	0.88	0.93 [0.89,0.96]	0.87 [0.8,0.92]
	Albright (2009)	0.93	0.96 [0.85,0.99]	0.96 [0.85,0.99]
	Daland et al. (2011)	0.88	0.86 [0.76,0.92]	0.71 [0.53,0.82]
Kendall’s τ_b	Scholes (1966)	0.71	0.77 [0.69,0.84]	0.65 [0.54,0.75]
	Albright (2009)	0.82	0.85 [0.70,0.99]	0.87 [0.80,0.95]
	Daland et al. (2011)	0.76	0.74 [0.66,0.81]	0.61 [0.50,0.73]

Table 6: Correlations of the negative exponentiated scores of the No-weights Hayes & Wilson Phonotactic Model trained on 1-frequency training data with results from the three experiments. Numbers in square brackets refer to the 95% confidence intervals. Red text indicates a significant difference.

3.6 Summary of model checks

The model checks gave us three results with respect to the Hayes & Wilson Phonotactic Model: (a) presence of gradience in the predictions doesn’t improve fit; (b) presence of gradience in the grammar (as constraint weights) itself doesn’t improve fit; (c) presence of gradience in the training data doesn’t improve fit.

To summarise, when we look more closely at model checks, we see that it is not clear that the gradience at multiple levels in the Hayes & Wilson Phonotactic Model is really useful. Crucially, the above model checks maintained the *all things kept equal* clause for model comparisons — therefore, the results can be appropriately inferred over.

At this point, it is important to pause and understand why the model fits are similar across such different implementations. To understand this, it is worth looking at the harmony scores for each of the models. In order to keep the discussion tractable here, I present a few selected harmony scores in Table 7. As can be seen for the different models, while the actual predicted scores are different, the different onset scores increase monotonically across all the models. Now, to be clear, while the monotonically increasing relationship is generally true, it is not perfect for the full set of stimuli across all the experiments.¹⁸ It is this general monotonic relationship between the scores that is ultimately responsible for the near identical fits across the different models tested. Consequently, what is of importance is not the weights of the constraints themselves, but whether the models produce an ordering relationship between the stimuli. And this can be done by the additive interaction of the

¹⁸The full set of predicted scores is in Appendix A

constraint violations themselves, without recourse to any weights associated with any of the constraints. Furthermore, given that the predictions for each of the modified models is highly linearly correlated with that of the original Hayes & Wilson Phonotactic Model predictions, they can be (roughly) linearly transformed from one set of predictions to another; *i.e.*, we don't need to know anything beyond the linear transform function that can be applied *across-the-board* to all the stimuli without knowing anything else of the stimuli to (approximately) figure out the values of one set of predictions from another. Therefore, the difference in predictions is somewhat like that of the difference between Fahrenheit and Celsius, where even if the actual values are different between the two temperature scales, values in one scale are not giving any additional information over the other, because they are perfectly linearly correlated. One should not be misled by the differences in the actual values from the models to infer that the models are showing us something meaningful, just as one is not to be misled by the fact that measurements of temperature are telling us different things just because our values are in Celsius vs. Fahrenheit.

Let's be more specific and take, for example, the case of the predictions of the No-weights model which appears to have a much smaller range of predicted values than the Original model. It is worth noting that the No-weights model assigns a weight of 1 to a violation and a weight of 0 to a non-violation — but this is an arbitrary decision. As long as every constraint gets the same numerical value for a violation and a non-violation, such a model would still be a no-weights model logically. That is, instead of mapping violated and unviolated to $\{1,0\}$, one could map it to any two constants $\{m,n\}$, and this is why the linear scale differences are not insightful in understanding model differences (see footnote 20 for related discussion.). For example, if one were to map every violation to a constant 3 instead 1 for the No-weights model, then the range of predicted values would be almost the same as that for the Original Hayes & Wilson Phonotactic Model model (see Appendix B for figures showing this point). Therefore, one should not be misled by the range of the predicted values in understanding the fit to human judgements, and we should instead depend on formal/mathematical measures of relationship such as Pearson's correlation or Kendall's τ_b .

	Onset	Orig.	EqFreq	No Wts	1-freq	EqFreq NoWts	1-Freq NoWts
Scholes (1966)	gɪ	0	0	0	0	0	0
	sk	0	0	0	0	0	0
	ft	-16.1	-15.1	-6	-5.8	-5	-3
	vz	-37.4	-36.3	-13	-10.4	-12	-5
Albright (2009)	pl	0	0	0	0	0	0
	pw	-5.2	-6.1	-1	-1.9	-1	-1
	pt	-14.8	-15.1	-5	-5.8	-5	-3
	bz	-33.3	-29.3	-11	-7.8	-10	-4
Daland et al. (2011)	bl	0	0	0	0	0	0
	fn	-12.3	-14.6	-4	-2.4	-4	-1
	ln	-26.4	-24.1	-8	-6.2	-8	-3
	ɹg	-44.0	-46.2	-13	-8.6	-14	-4

Table 7: Harmony scores for a few selected onsets for each of the models presented thus far

A second important issue is related to model (parameter) choices: what if the chosen parameters were different? In what follows, I compare the fits of the original model (parameter set 1) presented in Section 3.2 with two other parametric sets. One of the additional parameter sets (parameter set 2) allows for a maximum of a 100 constraints, has an O/E=0.8 (more liberal), allows for negation or complement sets, uses the original contrastive features used by Hayes and Wilson (2008), and has a window size of 2. the other additional parameter set (parameter set 3) has the same set of parameter decisions except O/E, which is set at 0.08 (more conservative). The correlations between the acceptability judgements and the MaxEnt scores from all these parameterisations are presented in Table 8. As can be observed, the parameterisation used in the current paper are quite comparable in fits to (and if anything, better than) the others, suggesting that perhaps the major work being done by the models is not dependent on the specific parameterisations themselves.

Correlation type	Data	Param. set 1	Param. set 2	Param. set 3
Pearson’s r	Scholes (1966)	0.88	0.88 [0.82,0.93]	0.70 [0.55,0.81]
	Albright (2009)	0.93	0.93 [0.75,0.98]	0.75 [0.27, 0.93]
	Daland et al. (2011)	0.88	0.84 [0.72,0.91]	0.66 [0.46,0.80]
Kendall’s τ_b	Scholes (1966)	0.70	0.70 [0.60,0.79]	0.70 [0.60,0.79]
	Albright (2009)	0.82	0.75 [0.57,0.92]	0.9 [0.81,0.99]
	Daland et al. (2011)	0.76	74 [0.68,0.81]	0.74 [0.67,0.82]

Table 8: Correlations between the average acceptability judgements and the negative exponentiated scores of the Hayes & Wilson Phonotactic Model with different parameterisations trained on the original frequency data. Numbers in square brackets refer to the 95% confidence intervals. Red text indicates a significant difference.

4 The minimal (categorical) phonotactic model

The previous section explored the utility (or lack thereof) of gradience at different levels with respect to the Hayes & Wilson Phonotactic Model. However, one should be careful in automatically extending the results to all computational models of gradience. As mentioned in the Introduction, such inferences are generally inappropriate as models come with many other auxiliary modelling/parametric choices that are usually not discussed or explored systematically. For example, the Hayes & Wilson Phonotactic Model has many more built in assumptions about how the constraints are learnt, under what regimes/criteria (Observed/-Expected metrics, simplicity, *etc.*). The presence of all of these modelling choices makes inference difficult for the theoretically interesting question of the need for gradience within the grammar.

To pursue the question, one really needs to start with a *minimal* model where the auxiliary assumptions are also simple (and preferably strongly motivated, independently). In what follows, I will propose a **minimal (categorical) phonotactic model** that uses features and segments. Such a model gets gradience in acceptabilities without any gradient generalisations. The model is based on a suggestion in Durvasula and Liter (2020). Durvasula and Liter (2020) used an Artificial Language Learning paradigm to study what phonotactic generalisations learners extracted from sequences of words that contained both continuant and voicing harmony (*e.g.*, *e.g.*, [fisu], [pita], [badi], ...) in the training phase. They observed that, when tested on novel consonant sequences that the participants didn’t hear in training, the participants only showed evidence of simple generalisations that voiced

consonants should co-occur with voiced consonants, and continuants should co-occur with continuants, but not the more complex generalisation that voicing and continuancy should both co-occur as in the training data. They suggest on p. 43 that “a more speculative possibility, one that takes a substantial inductive leap from our results, is that learners are only able to keep track of simple phonotactic generalizations, by which we mean generalizations that involve the precedence relationships between at most a single pair of features, segments, syllables, *etc.*. Therefore, the issue of complex vs. simple learned generalizations itself would vanish, as the learner simply cannot keep track of complex generalizations (of the relevant type). For example, learners might be able to keep track of precedence relationships such as [feature1]...[feature2], or [segment1]...[segment2], *etc.*, but not precedence relationships involving more than that, such as [feature1,feature3]...[feature2,feature3], or in the case of segments [segment1]...[segment2]...[segment3].” It is this model that I test in this section.

Ideally, in order to perfectly compare the results from the model against the results in the previous section, one should pursue this question within the context of the Hayes & Wilson Phonotactic Model itself. However, the focus here is to ask what is the *minimum* amount of architecture and auxiliary hypotheses necessary. To do this, we are forced to break from using the Hayes & Wilson Phonotactic Model as a framework for comparisons, as it comes baked in with many adjoining auxiliary assumptions that are not manipulable.

As a starting point for the comparison, it is worth thinking carefully about the space of possible grammars (and languages) with the Hayes & Wilson Phonotactic Model learner. With a 100 constraints that be of window size 1-3, there is a huge number of possible generalisations, 1-3.5 million in fact. Furthermore, the constraints can also have real-numbered weights. So, the space of possible grammars is technically uncountably infinite, thereby allowing for a lot of microparametric variation between participants based on the input they receive — a claim that requires a lot more careful experimentation to support than is currently reasonable (particularly because getting a proxy for each individual’s experience is quite difficult). With that as backdrop, it is worth asking what is the minimal model we can get away with in predicting human performance, and can we do it with minimal auxiliary assumptions (and preferably strongly motivated independently)?

The minimal (categorical) phonotactic model is a phonotactic model that contains all the observed single feature unigrams/bigrams and segment unigrams/bigrams in the learning corpus.

5. Example generalisations in the minimal (categorical) phonotactic grammar.

- *e.g.*, p r

feature unigrams: [+labial], [+voiceless], [+son], ...
feature bigrams: [+labial] - [+son], [+voiceless] - [+liquid], ...
segment unigrams: p, r
segment bigrams: #p, pr

Note: this is the minimal set of things needed by any grammatical model that has access to features and segments.

Within such a phonotactic model, any complex feature unigrams: $\begin{bmatrix} +labial \\ +voiceless \end{bmatrix}$ or complex feature bigrams: $\begin{bmatrix} +labial \\ +voiceless \end{bmatrix} \begin{bmatrix} +alveolar \\ +voiced \end{bmatrix}$, *etc.* are simply not learnable. So, the space of possible languages is quite small. With 25 consonants (including word boundary as a symbol) and 21 binary features, the unigram+bigram search space = 25 segments unigrams + 26*26 segment bigrams (including word boundaries) + 21*2 binary feature unigrams + (21*2+1)*(21*2+1) binary feature bigrams (including word boundaries) = 2592 generalisations.¹⁹ Furthermore, there are no weights to estimate.²⁰ So, this is a *finite* space of possibilities. An important consequence of such a model is that it immediately gives us a three-way distinction between *observed and grammatical* » *accidental gaps* » *ungrammatical*.²¹ I label the model the minimal (categorical) phonotactic model because it is the smallest model that achieves the desideratum of the grammatical knowledge itself predicting this trichotomy. With accidental gaps, while the featural combinations might all be present, the actual segment sequence will still be missing from the phonotactic grammar.²²

Next, we need (preferably) independently motivated auxiliary assumptions to get the grammar to make contact with acceptability judgements. Essentially, we need a linking hypothesis about *performance*. The performance related linking hypothesis is as follows: for

¹⁹With trigrams: it increases to 99,675 generalisations

²⁰Of course, one could claim that there are weights even in this model, and that they are boolean; however, the term “weight” is often associated with non-boolean (likely, natural or real numbered) values in Linguistics, and therefore I have maintained that custom here in phrasing here.

²¹Note, the model does make the claim that all observed sequences are grammatical. However, note that some “exotic” sequences were removed from the training data. I return to this important issue in the Conclusion.

²²It is perhaps possible for one to model the intuitions in a harmonic grammar (Legendre, Miyata, and Smolensky 1990a,b; Smolensky and Legendre 2006), but the intuitions of additive effects of ungrammaticality precede modern harmonic grammars in Phonology. For example, similar additivity was argued for in Chomsky (1986a), where no harmonic grammar was intended.

each nonsense test word, one counts the number of featural and segment unigrams/bigrams that are not in the grammar (Durvasula 2020; Kostyszyn and Heinz 2021), and then applies add-1 Laplace correction (to avoid the problem of zero counts).²³ Finally, a $\log(\text{corrected count})$ is calculated. This last step is of course a very well-motivated general principle in cognition, namely the *Weber-Fechner law* (Sun et al. 2012), as applied to numerical cognition — where the mental number line is seen to be logarithmic instead of linear (Dehaene 2003; Dehaene et al. 2008). There are no additional (or free) parameters to fit to the data, and therefore there is no wiggle room in the predictions. Note, the ngrams learnt are not free parameters: the usual meaning of the term “free parameter” is to indicate a parameter that can be adjusted to improve data fit *post hoc*. Here, the relevant data is the set of acceptability judgments, and the model is the set of ngrams learnt based on the training data. The ngrams in the model are thus not free parameters because they are not adjusted based on fit to acceptability judgements.

4.1 Results

In this sub-section, I compare the predictions of the minimal (categorical) phonotactic model with the acceptability judgements from all three sets of experimental results under consideration in this paper, and compare such model fits against those from the original Hayes & Wilson Phonotactic Model that were presented in the previous section. I will compare both the model scores themselves and the negative exponentiated version of the scores, so that the comparison is maximally similar to the modelling choices of the Hayes & Wilson Phonotactic Model. Again, the attempt is to, as much as possible, compare apples to apples.

As can be seen in Figures 11-10, the minimal (categorical) phonotactic model fits the human judgements quite well.

²³Note, this correction is not independently psychologically motivated, and therefore it can and should be debated. But, I will for current purposes go along with it, since some sort of 0 frequency correction is standard in computational modelling. We can however weakly motivate this choice as piggy-backing on the Weber-Fechner law. If, indeed, some sort of logarithmic transform is done over counts, then one needs non-zero counts, as $\log(\text{zero})$ is not defined. And since counts are integer-valued, the smallest addition one can imagine is 1 to solve the problem — this motivates the 1-Laplace correction.

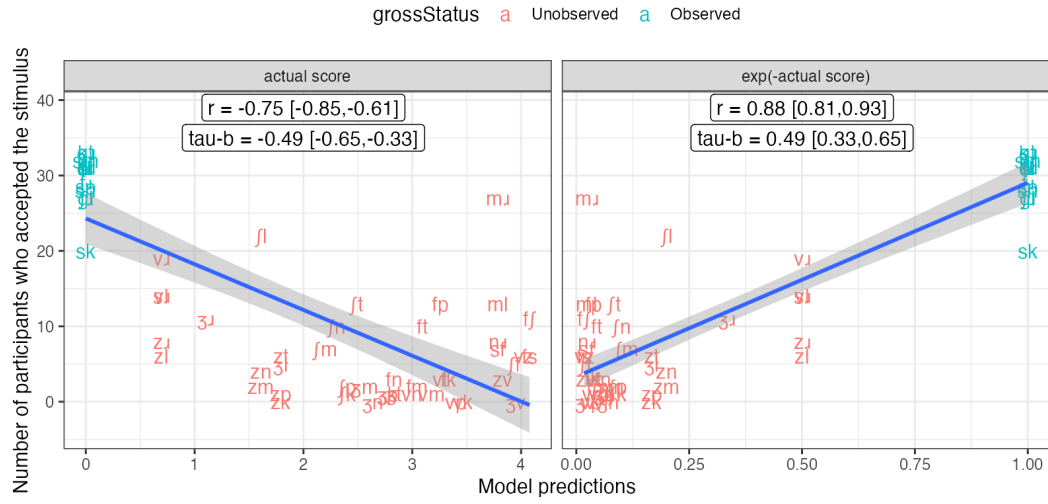


Figure 10: Comparison of the minimal (categorical) phonotactic model predictions to human judgements for the Scholes (1966) data. Numbers in square brackets refer to the 95% confidence intervals.

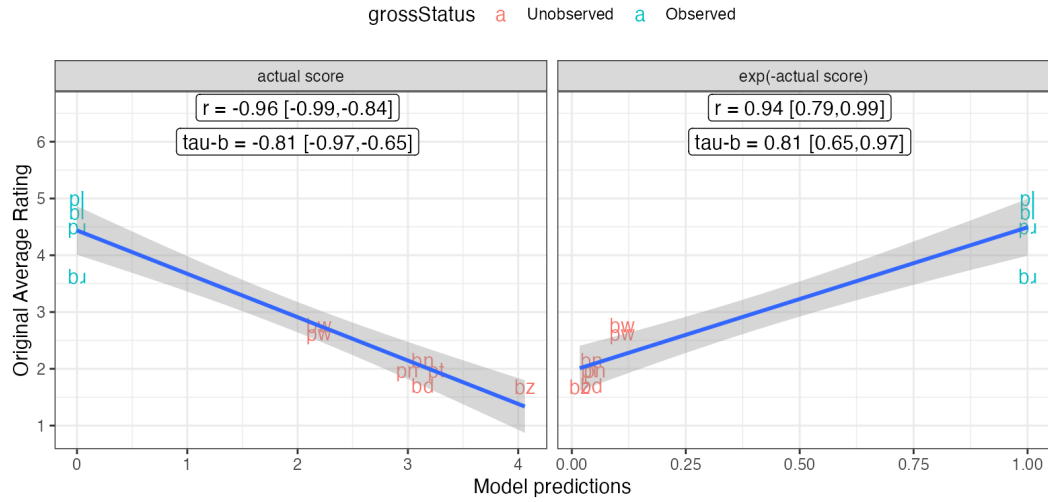


Figure 11: Comparison of the minimal (categorical) phonotactic model predictions to human judgements for the Albright (2007) data. Numbers in square brackets refer to the 95% confidence intervals.

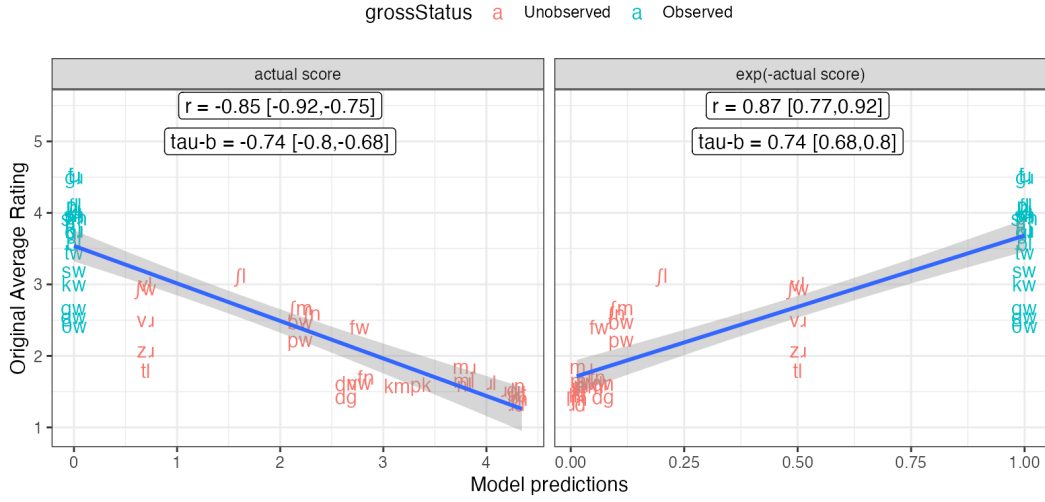


Figure 12: Comparison of the minimal (categorical) phonotactic model predictions to human judgements for the Daland et al. (2011) data. Numbers in square brackets refer to the 95% confidence intervals.

When we compare how well the minimal (categorical) phonotactic model fits human acceptability judgements when compared to that of the original Hayes & Wilson Phonotactic Model, it performs remarkably well. In fact, the minimal (categorical) phonotactic model does better when looking at the raw Pearson’s r values for either the raw or the negative exponentiated scores. The Hayes & Wilson Phonotactic Model does better only in the case of the Scholes’s (1966) data, for Kendall’s τ_b . So, the only comparison where the Hayes & Wilson Phonotactic Model does better is one where the measure of fit collapses a crucial distinction made by the model. This is therefore difficult to see as strong evidence for the model. Furthermore, the fact that the other experimental results don’t show a similar result is more of a reason to think that, in general, the minimal (categorical) phonotactic model does as well as the Hayes & Wilson Phonotactic Model.

Correlation	Data	Hayes and Wilson (2008) (raw/exp)	Minimal model (raw)	Minimal model (exp)
Pearson’s r	Scholes (1966)	-0.75 / 0.88	-0.75 [-0.85,-0.61]	0.88 [0.81,0.93]
	Albright (2007)	-0.83 / 0.94	-0.96 [-0.99,-0.84]	0.94 [0.79,0.99]
	Daland et al. (2011)	-0.73/0.86	-0.85 [-0.92,-0.73]	0.87 [0.77,0.92]
Kendall’s τ_b	Scholes (1966)	0.71	-0.49 [-0.65,-0.33]	0.49 [0.65,0.33]
	Albright (2007)	0.82	-0.81 [-0.97,-0.65]	0.81 [0.65,0.97]
	Daland et al. (2011)	0.76	-0.74 [-0.80,-0.68]	0.74 [0.68,0.80]

Table 9: Summarising the correlations for the different models. Numbers in square brackets refer to the 95% confidence intervals. Red text indicates a significant difference.

An interesting aspect of the performance of the minimal (categorical) phonotactic model is that aside from the Pearson’s r value for the Scholes (1966) data, the negative exponentiation doesn’t really change the fits to human judgements much. If this result holds up in the future, it would suggest that the gradience in this particular model’s predictions might be picking up something useful.

To summarise, despite it being the *minimal* model, the model generally does at least as well as the Hayes & Wilson Phonotactic Model which has an incredibly rich architecture.

5 Conclusion

In this paper, I presented modelling results and model checks related to the Hayes & Wilson Phonotactic Model and modelling related to a new **minimal (categorical) phonotactic model**. With respect to the Hayes & Wilson Phonotactic Model, though it has been touted as evidence of gradient phonotactic knowledge, it was observed in a series of model checks that claims of evidence of gradience are not sustainable, at least for the case of English word-initial consonant sequences — the very case that was used to show evidence for the model in the first place. There was little to no evidence that gradience in the predictions or in the constraint weights or in the training data appreciably improved the model fit to human judgements. The results obtained are reminiscent of the results in Dai (2025), Gorman (2013), and Kostyszyn and Heinz (2021), who all argued that the well-specified categorical models do just as well as gradient models.²⁴ However, the focus in the current

²⁴Albeit, note that all these grammars were both intensionally and extensionally categorical — a point I will return to shortly.

paper was to show that the touted performance of the gradient model (in our case the Hayes & Wilson Phonotactic Model) itself doesn't come from any gradient in the predictions or grammar or training data.

In previous work arguing for categorical models, the gradient observed in the judgements was largely chalked up to (although, not discussed elaborately) noise. A useful extension of this idea is to interpret "noise" as a cover term for various unidentified non-grammatical sources. In fact, one such non-grammatical source is the influence of perception in judgements. As Kahng and Durvasula (2023) show, perceptual gradient has a sizeable effect on the ratings of nonwords, and therefore, should be factored in in interpreting gradient in acceptability judgements. However, even if one were to entertain the possibility that some of the gradient in the acceptability judgements stems from the gradient in the predictions of the grammar, it doesn't follow that the grammar itself has *gradient knowledge* (or constraints in our case), *i.e.*, even if there is a grammar is *extensionally* gradient, one shouldn't automatically assume that it is *intensionally* gradient. To make this argument clear, I proposed a minimal (categorical) phonotactic model with very few auxiliary assumptions and no free parameters at all. This latter model generally fit human judgements at least as well as the more complex Hayes & Wilson Phonotactic Model. Furthermore, the model predicts some amount of gradient in judgements along with getting the important distinction between *observed and grammatical* » *accidental* » *ungrammatical*. In short, a grammar can lead to systematic gradient acceptability judgements, yet the knowledge (or constraints) within the grammar themselves are categorical.

The results in the paper suggest that likely much more can be learnt from the nature of the constraints/rules and the representations used in the grammar, than the issue of gradient itself (see (Rawski 2021; Swanson, Heinz, and Rawski [accepted](#)) for related discussion). That is, the results to my eye suggest a return to traditional phonological concerns, largely because they are likely to bear (bigger) fruit. For example, the minimal (categorical) phonotactic model allows for only single feature bigrams (or n-grams) and segment bi-grams — this is a rather limited constraint space. Note, the model was not only found to be predictive in this paper, but it was also found to be useful in understanding the artificial language results in Durvasula and Liter (2020), where they observed that participants are able to learn multiple single-feature bigram generalisations, but not more complex multi-feature bigram generalisations. However, it is worth pursuing a project where the constraint space (constraint window size and constraint depth) is systematically modified, while keeping everything else constant, to see if there is an improvement in model performance when

compared against human judgements. Given how well the minimal (categorical) phonotactic model actually performs in this regards, perhaps the improvement will be incremental, in which case it is important to ask why the increase in constraint/rule space leads to only incremental improvements and if we should be deriving the increased model fit through other non-grammatical factors.

In line with the general advice in this paper that we should be careful in inferring aspects of the grammar from computational models, one can turn a critical eye even towards the minimal (categorical) phonotactic model that I proposed, and try to decompose the effects into an even simpler model, where there are only single feature unigrams/bigrams in the phonotactic grammar, and segmental generalisation fall out of extra-grammatical systems. Note, the juxtaposition of both segmental and featural representations within a grammatical model is quite weird if segments are seen as epiphenomenal constructs based on features. Therefore, we should be probing this issue further. Perhaps, the segmental knowledge is actually from orthography/orthotactic knowledge and not phonological knowledge *per se*. There is by now a fair bit of evidence that there are orthographic effects in speech production, perception and even in loanword patterns (Grippando 2019; Stoehr and Martin 2022; Vendelin and Peperkamp 2006; Warner et al. 2004) — all showing that orthographic and orthotactic knowledge affect performance, which is thought to be multi-factorial anyway. So, though currently just a speculation, it wouldn't be surprising at all if orthotactic knowledge is the ultimate source for the segmental effects in acceptability judgements. While fairly specific orthotactic models are required to make progress, it does highlight the main point of this paper that a good computational model fit should not be directly interpreted as evidence of specific theoretical constructs (included within the computational model) without proper model checking/inspection and understanding non-grammatical factors, particularly when it means we are enlarging the space of possibilities for the grammar.

Furthermore, here is potentially where the no-phonotactic grammar view that Reiss (2017) espouses does have a prediction to make. If that view is right, and if segmental (not featural) knowledge is what is driving judgements, and is itself from orthotactic knowledge (and not true phonological knowledge, *per se*), then we should see a difference between literate and illiterate speakers. While such tests are quite minimal in modern linguistic research, there is at least somebody of work discussing similar questions related to literacy (Linell 1979), but see Kazanina, Bowers, and Idsardi (2017) for an alternative view of how orthography might only indirectly affect performance by essentially making speakers more meta-linguistically aware. Furthermore, the view would still have to be reconciled with

results of pre-literate children who show clear evidence of sensitivity to phonotactic patterns (Friederici and Wessels 1993; Jusczyk et al. 1993; Sebastián-Gallés and Bosch 2002). However, there is a larger point here that if we are interested in learning about the underlying competence/grammatical system, we need to be as critical as possible, and only allow new/more constructs into the system if they are theoretically unavoidable and can be justified through repeated experimentation and analysis — the simple fact of just accounting for the data is far from sufficient to make the case. Furthermore, theorists and computational researchers have to take performance factors seriously and incorporate such effects in the modelling or devise experiments that somehow sidestep such concerns or come up with the relevant *gedanken* experiments that go beyond the concerns to make an appropriate inference. While others might disagree, this has been the model for the rest of the natural sciences from my lens, and therefore, I see benefit in emulating that approach - not because there is a guarantee, but because there is no better way that I am aware of. Either way, just acknowledging non-grammatical factors and then simply ignoring such factors and fitting computational models instantiating specific grammatical views seems somewhat inappropriate²⁵

Finally, and related to the last point, many computational models, incorporating either gradient and categorical grammars, include purely segmental representations (Dai 2025; Gorman 2013; Mayer *in press*, amongst other). It is worth noting, in line with the previous point that this is a remarkable departure from modern phonological representations, which are almost completely feature- or autosegment- or element-based (Chomsky and Halle 1968; Goldsmith 1976; Kaye, Lowenstamm, and Vergnaud 1985). While segmental representations are often used as a simplification, they are not innocent simplifications. Segmental and featural models have very different spaces of possibilities and can lead to substantial differences in the behaviour of the grammars, depending on how they are implemented. For example, Albright (2009) observed that the two types of grammars allow for different degrees of fit to observed judgement data. For the same reason mentioned in the Introduction, related to the difficult of making inferences about the underlying grammar when other decisions are different, it is difficult to know how much of the modelling success crucially depends on the representations chosen; therefore, it is difficult to interpret such computational models as saying anything about phonological grammars as conceived in current day terms. As with the previous point, it would be optimal to derive the segmental effects ultimately from

²⁵There is a distinct fear that simply ignoring non-grammatical factors will not allow us to make progress and the multitude of computational models will just increase without resolution. As a result, we as a field are stuck between playing whack-a-mole and chasing our own tails — both entertaining, neither productive.

non-grammatical sources (including orthotactics), if one is committed to features or autosegments or elements as the true phonological primitives. Alternatively, one can discard those representations in favour of segmental representations as the true phonological primitives.

5.1 Additional issues, clarifications and speculations

There are five more related issues worth elaborating on. First, sometimes computational models come with *free parameters*, parameters that are simply fit to the very same data that the models are trying to predict. For example, the Hayes & Wilson Phonotactic Model comparisons have often been accompanied with a separate *temperature* parameter, which is a non-linear scaling parameter used on the predictions that allows a better fit to human judgement data. Importantly, an appropriate temperature parameter value can be found to both *decrease* or *increase* model fit, and therefore the use of such free parameters actually makes it harder to understand which aspects of the model are doing the grunt work. Furthermore, when there is variance in a measure, one can always improve the fit on some data through over-fitting, and going beyond what is truly justifiable from the data. The temperature parameter could therefore result in simply over-fitting the results. This is the main reason to avoid the temperature parameter in model comparisons. In contrast, the minimal (categorical) phonotactic model proposed is not meant to be used with any free parameters.

Second, the results in this paper should not be seen as a falsification of phonotactic grammars incorporating gradience, but as a foray into *understanding* where the gradience in acceptability judgements comes from. Future work might very well reveal that there are other domains of phonotactic judgements where gradient grammars fit the judgement data better. However, in line with the note of caution throughout, this should not be interpreted as evidence of gradient phonotactic grammars without extensive model checking and replication. Furthermore, if some domains show that gradient phonotactic models fit the judgement data better, and other domains show that categorical models are better, this is chance to understand *why* that is the case.

For example, recently, Mayer ([in press](#)) presented results from a very careful study that compared model performance against native Turkish speakers' judgements of CVCVC non-words. Specifically, they compared minimally different models (very much along the lines advocated by the current article) incorporating bigram information over vowel sequences (strict locality was not necessary for the calculation). The gradient model incorporated bigram frequency information, while the categorical models only had access to the pres-

ence/absence of the bigram. They observed that the gradient model fit the human judgements better. There are three issues that we need to put aside even though they are extremely relevant: (a) that probabilistic grammars that model acceptability as a product of bigram probabilities make a deeply problematic prediction that longer words will be more ungrammatical (see Alves 2023; Coleman and Pierrehumbert 1997; Miller 1957, for discussion); (b) that it is not clear that the gradient bigram model that Mayer (in press) used actually fit the human judgements better than the other models they considered, since confidence intervals for the fit measures were not provided. I put the above issues aside though they are quite relevant, because even if one grants the above are not problems, there is still an experimental issue to consider. Given the results in the current article that many categorical phonotactic models do at least as well as gradient phonotactic models and the fact that Daland et al. (2011) found a bigram model did worse than the Hayes & Wilson Phonotactic Model for their English results²⁶, one would expect that categorical models to do better than corresponding gradient models. But, then, how does one square the results from the Turkish judgments who found gradient phonotactic models to be better than categorical equivalents? Note, the stimuli in Mayer (in press) involve CVCVC words, where almost all the vowel sequences are observed in the training corpus. Table 10 shows the observed V-V frequencies in the Turkish Electronic Living Lexicon (Inkelas et al. 2000).²⁷ This is in stark contrast to the stimuli in Albright (2007), Daland et al. (2011), and Scholes (1966), where a fair portion of the nonwords contained completely *unobserved* sequences. So, the task in Mayer’s (in press) experiment is substantially different from the one participants performed in the experiments used in the current paper. The former involved almost completely observed segment sequences, while the latter didn’t.²⁸ We know that in other domains, participants often re-interpret tasks and provide inappropriate evidence of specific claims (Armstrong, Gleitman, and Gleitman 1983; Firestone and Scholl 2016; Schütze 2005). The participant in Mayer’s (in press) experiment is perhaps forced to re-interpret the task as deciding between which observed sequence is a *better* word of Turkish — when the grammar is silent on the distinction, the participant is forced to depend on lexical similarity

²⁶Albeit, it is not clear from their results that the Hayes & Wilson Phonotactic Model fit was statistically significantly better than that of the bigram model.

²⁷I extracted these frequency counts from the file labelled `MASTER.db.txt` in the online database available at <https://linguistics.berkeley.edu/TELL/>.

²⁸This distinction is akin to asking a participant to judge sentences as acceptable when all the sentence frames in the experiment are observed (and thus likely grammatical), versus asking them to judge observed vs. (likely) ungrammatical sentences. The tasks are quite substantially different at face value.

effects, perhaps.²⁹ If so, it is not surprising that there is more of an influence of gradient information in Mayer’s (in press) results. Note, this is a possibility that deserves further empirical probing. For now, I just raise it to point out that the real challenge is in figuring out how to interpret conflicting fits to human judgements from different phonotactic domains/questions.

	a	e	i	o	u	y	ø	ʊ
a	4,186	1,023	1,463	387	648	113	52	2,014
e	1,424	2,024	2,036	194	258	104	27	44
i	1,035	780	1,165	219	39	48	21	50
o	981	183	163	182	557	29	25	11
u	1,120	165	103	32	698	20	12	18
y	418	1,063	119	24	92	589	18	14
ø	49	442	14	25	8	256	17	8
ʊ	583	63	51	18	8	6	10	659

Table 10: V_1 - V_2 frequencies in the TELL corpus

Third, in this paper, to keep as many aspects of the model comparison identical as possible, I maintained Hayes and Wilson’s (2008) original modelling decision of limiting the training word-initial consonant sequences to a subset of what is actually observed in the CMU dictionary. As mentioned in Section 2.1, Hayes and Wilson’s (2008) rationale was that they wanted to remove any consonant sequences that they judged to be “exotic”. It is for this reason that both the Hayes & Wilson Phonotactic Model and the minimal (categorical) phonotactic model probed in this paper are not learning models *per se*, but grammatical models. From a learner’s perspective, the relevant question is of course, one of identifying the exotic consonant sequences in the first place. Here, one could make an argument that some sort of lexical frequency information is valuable to identify the set of sequences to learn from.

²⁹The phrase “lexical similarity effect” is to be contrasted with an effect driven by a phonotactic grammar. By the former, I mean an effect that arises with a direct comparison to the lexicon, as in Chomsky and Halle (1968) and Greenberg and Jenkins (1964, amongst others). By the latter, I mean an effect that is driven by (a comparison to) some independent phonotactic knowledge, akin to redundancy rules or Morpheme Structure Constraints or a Hayes & Wilson Phonotactic Model or a bigram model (to the extent that the latter is thought to be a separate phonotactic grammar). Note, the above understanding of an independent phonotactic grammar is also present in Hayes and Wilson (2008), and one of the accounts of phonological acceptability judgements discussed by Chomsky and Halle (1968) that was cashed out in terms of redundancy rules.

This is exactly the tack taken by Dai (2025).³⁰ However, the issue of using some frequency information (if at all) to identify the sequences to generalise over should be kept separate from the issue of the nature of the grammar itself that is learned from such a set of inputs or the issue of whether frequency information is used once such a set of sequences has been identified. Therefore, it is possible that frequency information is used for subsetting to the relevant set of stimuli to learn over, but not for the sake of utilising the observed frequencies to learn the grammar *per se*. Even here, there is a second alternative — one needs to look at the words such “exotic” consonant sequences appear in. If they appear in words that are unlikely to be experienced by a learner within a certain critical (age) window (Lenneberg 1967), then perhaps the learner fails to include them in the set of sequences to learn over. For example, take the case of the word-initial consonant sequence [sf], the CMU dictionary has a total of 12 words with that sequence.³¹ The entire Childe-North American corpus (MacWhinney 2000) including conversations between parents/investigators and about 1117 children (under 11 years of age, and most under 4 years of age) has a total of 5 tokens of the words containing the sequence (4 <sphere>, 1 <Sphinx>). Just as a comparison to get an upper-bound, there were 375,846 tokens of the word <the>.³² If so, then there is no need for frequency information even in identifying the relevant set to learn over. For this possibility, one needs to look at the actual input that children get within the first few years of their lives.

Fourth, it is important to clarify that, in line with Chomsky and Halle (1968), Gorman (2013), and Hayes and Wilson (2008), whatever our theory of phonotactics is, it is ultimately separate from our theory of phonological alternations, since not all alternations solve phonotactic problems, and some alternations are even “crazy” (Bach and Harms 1972; Buckley 2000; Scheer 2009). In this paper, I have focussed on the issue of phonotactic knowledge alone. Furthermore, how the model applies to (if at all) our understanding of loanword adaptations is a question I leave for further research. However, it is worth pointing out that the work on loanword phonology is complicated by the fact that it is often claimed to be generated by grammars that are different from the L1 grammars (Kang 2011), and loanword patterns are affected by factors well-beyond the phonotactics of the borrowing language, such as orthography, mismatches in the phonetics between source and borrowing languages,

³⁰This is an approach that has been espoused in other work on the productivity of linguistic generalisations (Yang 2016).

³¹The full list is as follows: <sfernice, sferra, sferrazza, sfed, sforza, sfuzzi, sphar, sphere, spheres, spherical, spheroid, sphinx>. Many of these words are unknown even to me, let alone an infant.

³²I extracted these counts using the R library, `chldesr` (Sanchez et al. 2019), which used the 2021.1 version of the Childe database.

... (Kang 2011; Smith 2006, to appear) — this makes using loanword patterns as evidence of phonotactic grammars, tenuous, at best. For this reason, instead of testing on loanword patterns, it is perhaps better to pursue a line of research along of Caplan, Kodner, and Yang (2020) to see if phonotactic models, including the minimal (categorical) phonotactic model, make the right predictions about the kinds of words that enter the language.

Finally, some of the aspects I touched on in this paper are awfully similar to the *bounded rationality* work by Gerd Gigerenzer and colleagues (Mousavi and Gigerenzer 2014), where they use the distinction between *risk* and *uncertainty* (Knight 1921) — *risk* refers to the case where the full sample space and associated probabilities are known or fully estimable, as in games of chance, while *uncertainty* refers to more real world conditions where the sample space is not fully specified, or therefore probabilities are difficult (and perhaps impossible in a technical sense) to estimate. Gerd Gigerenzer and colleagues have argued that complex/gradient optimisation perform better in the former case of *risk*. However, simplicity and heuristic thinking perform better in the latter case of *uncertainty*, where estimation is less accurate, particularly when the predictive uncertainty of the problem and the number of possibilities is large, but the size of the learning sample is small (Gigerenzer 2008). The case of phonotactics in natural language clearly falls in the *uncertainty* camp, and therefore it is quite heartening to see the convergence between the results in the current paper and those of general cognition related to decision making.

6 Acknowledgements

A Harmony scores for all the Hayes & Wilson Phonotactic Model model alternatives

Table 11: The harmony scores of Hayes & Wilson Phonotactic Model compared to those for the different model checks for the Albright (2009) data

IPA	Original	NoWeights	EqFreq	OneFreq	EqFreqNoWeights	OneFreqNoWeights
pl	0	0	0	0	0	0
bl	0	0	0	0	0	0
pɪ	0	0	0	0	0	0

bl	0	0	0	0	0	0
bw	-5.162	-1	-6.125	-1.931	-1	-1
bn	-10.647	-3	-6.125	-1.931	-1	-1
pn	-14.796	-5	-15.146	-5.76	-5	-3
pt	-14.796	-5	-15.146	-5.76	-5	-3
bd	-20.757	-8	-16.684	-5.76	-7	-3
bz	-26.065	-9	-24.169	-5.76	-9	-3
bl	-33.276	-11	-29.251	-7.828	-10	-4

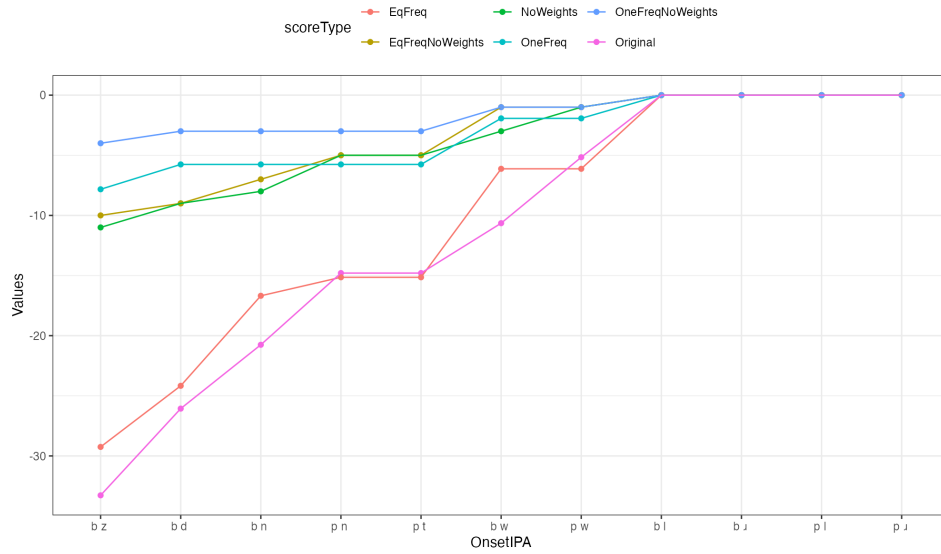


Figure 13: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the different model checks for the Albright (2009) data

Table 12: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the different model checks for the Daland et al. (2011) data

IPA	Original	NoWeights	EqFreq	OneFreq	EqFreqNoWeights	OneFreqNoWeights
bl	0	0	0	0	0	0
bl	0	0	0	0	0	0
dl	0	0	0	0	0	0

fl	0	0	0	0	0	0
fr	0	0	0	0	0	0
gl	0	0	0	0	0	0
gr	0	0	0	0	0	0
kl	0	0	0	0	0	0
kr	0	0	0	0	0	0
kw	0	0	0	0	0	0
pl	0	0	0	0	0	0
pr	0	0	0	0	0	0
sm	0	0	0	0	0	0
sn	0	0	0	0	0	0
sw	0	0	0	0	0	0
fr	0	0	0	0	0	0
tr	0	0	0	0	0	0
tw	0	0	0	0	0	0
dw	-2.854	-1	0	0	0	0
gw	-2.854	-1	0	0	0	0
w	-3.569	-1	0	0	0	0
fl	-4.177	-1	-4.8	0	-1	0
pw	-4.927	-1	-6.125	0	-1	0
fw	-4.927	-1	-6.125	0	-1	0
tl	-5.128	-1	-5.783	0	-1	0
ml	-5.322	-1	-6.575	0	-1	0
mr	-5.322	-1	-6.575	0	-1	0
vl	-5.463	-2	-7.021	-2.615	-2	-1
vr	-5.463	-2	-7.021	-2.615	-2	-1
bw	-7.781	-2	-6.125	0	-1	0
fw	-8.496	-2	-6.125	0	-1	0
km	-9.456	-4	-9.02	-5.96	-4	-3
nl	-10.45	-2	-13.492	0	-3	0
fm	-12.262	-4	-14.596	-2.364	-4	-1
fn	-12.262	-4	-14.596	-2.364	-4	-1
zr	-13.229	-4	-12.434	-2.615	-3	-1
pk	-14.383	-5	-15.146	-5.96	-5	-3

dn	-14.725	-5	-13.805	-3.596	-6	-2
vw	-16.813	-5	-13.147	-2.615	-3	-1
ɹl	-17.451	-4	-22.017	-2.615	-5	-1
fn	-17.952	-6	-15.146	-5.96	-5	-3
dg	-20.056	-6	-21.29	-3.596	-8	-2
lm	-26.44	-8	-24.105	-6.21	-8	-3
ln	-26.44	-8	-24.105	-6.21	-8	-3
lt	-26.44	-8	-24.105	-6.21	-8	-3
ɹm	-38.702	-12	-38.701	-8.574	-12	-4
ɹd	-44.034	-13	-46.187	-8.574	-14	-4
ɹg	-44.034	-13	-46.187	-8.574	-14	-4

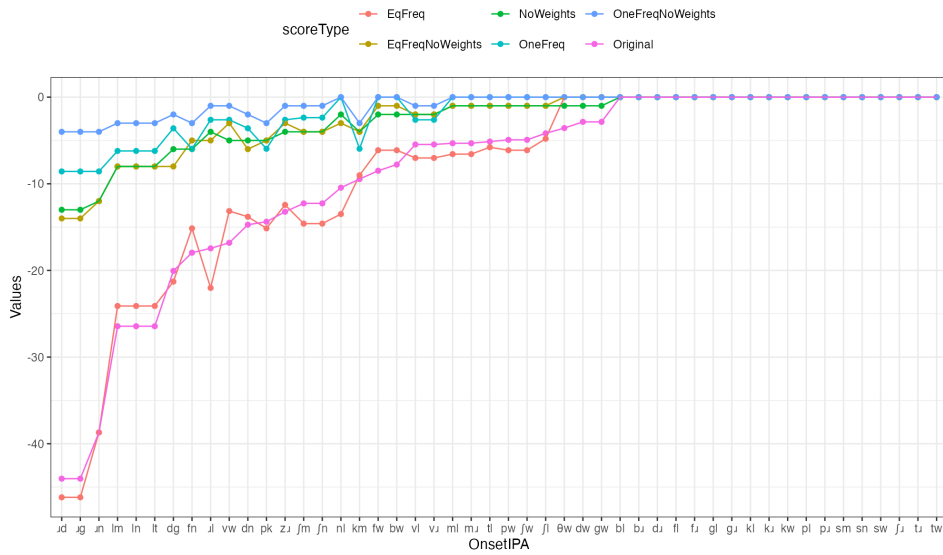


Figure 14: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the different model checks for the Daland et al. (2011) data

Table 13: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the different model checks for the Scholes (1966) data

IPA	Original	OriginalRepeated	EqFreq	NoWeights	OneFreq	EqFreqNoWeights	OneFreqNoWeights
-----	----------	------------------	--------	-----------	---------	-----------------	------------------

gɪ	0	0	0	0	0	0	0
kɪ	0	0	0	0	0	0	0
st	0	0	0	0	0	0	0
sm	0	0	0	0	0	0	0
pɪ	0	0	0	0	0	0	0
sl	0	0	0	0	0	0	0
fl	0	0	0	0	0	0	0
bl	0	0	0	0	0	0	0
dɪ	0	0	0	0	0	0	0
tɪ	0	0	0	0	0	0	0
fɪ	0	0	0	0	0	0	0
sp	0	0	0	0	0	0	0
sn	0	0	0	0	0	0	0
ʃɪ	0	0	0	0	0	0	0
gl	0	0	0	0	0	0	0
sk	0	0	0	0	0	0	0
ʃl	-4.177	-4.177	-4.8	-1	0	-1	0
sɪ	-4.765	-5.078	-5.413	-1	0	-1	0
vɪ	-4.774	-5.463	-7.021	-1	-2.599	-2	-1
vl	-4.774	-5.463	-7.021	-1	-2.599	-2	-1
mɪ	-5.257	-5.322	-6.575	-1	0	-1	0
ml	-5.257	-5.322	-6.575	-1	0	-1	0
nɪ	-5.257	-5.322	-6.575	-1	0	-1	0
zl	-7.47	-8.151	-8.155	-2	-2.599	-3	-1
ff	-8.514	-7.961	-9.723	-3	-5.897	-3	-3
sf	-9.234	-9.616	-10.495	-2	-2.068	-2	-1
ʒɪ	-9.541	-12.2	-11.811	-3	-2.599	-3	-1
zt	-10.674	-11.45	-9.694	-4	-2.599	-5	-1
zn	-10.674	-11.45	-9.694	-4	-2.599	-5	-1
zm	-10.674	-11.45	-9.694	-4	-2.599	-5	-1
zp	-10.674	-11.45	-9.694	-4	-2.599	-5	-1
zk	-10.674	-11.45	-9.694	-4	-2.599	-5	-1
zɪ	-12.235	-13.229	-12.434	-3	-2.599	-3	-1
ʃt	-12.237	-12.262	-14.596	-4	-3.722	-4	-2

fn	-12.237	-12.262	-14.596	-4	-3.722	-4	-2
fm	-12.237	-12.262	-14.596	-4	-3.722	-4	-2
fp	-12.237	-12.262	-14.596	-4	-3.722	-4	-2
fk	-12.237	-12.262	-14.596	-4	-3.722	-4	-2
gl	-13.718	-16.376	-17.744	-4	-2.599	-5	-1
fp	-16.094	-17.952	-15.146	-6	-5.76	-5	-3
ft	-16.094	-17.952	-15.146	-6	-5.76	-5	-3
fn	-16.094	-17.952	-15.146	-6	-5.76	-5	-3
fk	-16.094	-17.952	-15.146	-6	-5.76	-5	-3
fm	-16.094	-17.952	-15.146	-6	-5.76	-5	-3
ff	-16.706	-16.8	-19.678	-5	-5.79	-5	-3
fs	-20.563	-22.49	-20.228	-7	-7.828	-6	-4
vt	-24.073	-26.714	-23.705	-9	-8.36	-9	-4
vm	-24.073	-26.714	-23.705	-9	-8.36	-9	-4
vn	-24.073	-26.714	-23.705	-9	-8.36	-9	-4
vp	-24.073	-26.714	-23.705	-9	-8.36	-9	-4
vk	-24.073	-26.714	-23.705	-9	-8.36	-9	-4
zm	-24.982	-27.761	-29.079	-9	-6.322	-10	-3
zp	-24.982	-27.761	-29.079	-9	-6.322	-10	-3
zt	-24.982	-27.761	-29.079	-9	-6.322	-10	-3
zk	-24.982	-27.761	-29.079	-9	-6.322	-10	-3
zn	-24.982	-27.761	-29.079	-9	-6.322	-10	-3
zv	-26.066	-26.398	-27.673	-8	-4.667	-9	-2
zv	-35.609	-37.631	-41.647	-12	-8.39	-13	-4
vz	-37.395	-39.272	-36.272	-13	-10.428	-12	-5

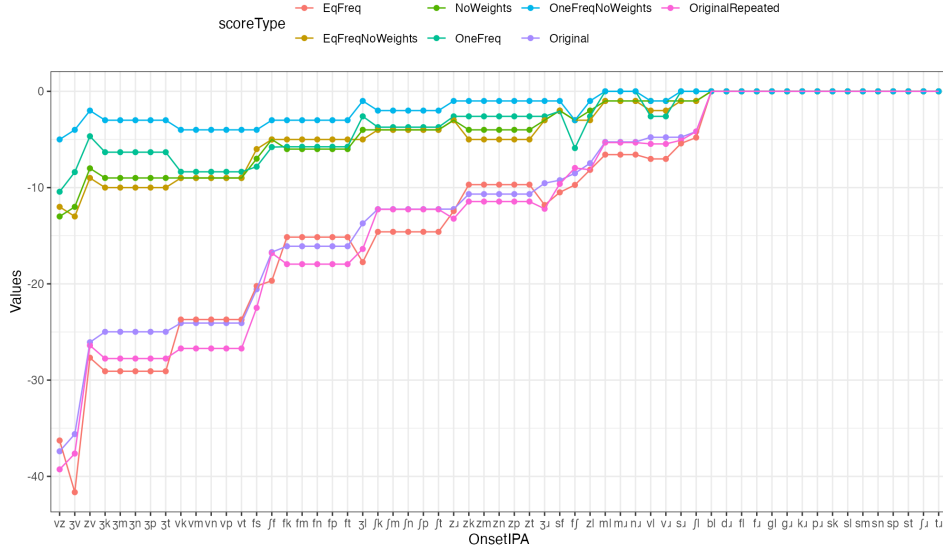


Figure 15: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the different model checks for the Scholes (1966) data

B A No-weights model with a different range of predicted values

A reviewer worries that there are substantive differences between the models because the ranges of the predicted models are substantially difference. However, this is the issue of scale-dependence alluded to earlier.

Let’s take, for example, the case of the predictions of the No-weights model which appears to have a much smaller range of predicted values than the Original model. It is worth noting that the No-weights model assigns a weight of 1 to a violation and a weight of 0 to a non-violation — but this is an arbitrary decision. As long as every constraint gets the same numerical value for a violation and a non-violation, such a model would still be a categorical no-weights model logically, as each constraint maps to one of only two possible states. That is, instead of mapping a violation and a non-violation to $\{1,0\}$, one could map it to any two constant $\{m,n\}$, and this is one reason why the linear scale differences are not useful in insightful in understanding model differences (see footnote 20 for related discussion). For example, if one were to map every violation to a constant 3 instead of 1 for the No-weights model, then the range of predicted values would be almost the same as that for the Original

Hayes & Wilson Phonotatic Model model, as can be seen in the figures below. Therefore, one should not be misled by the range of the predicted values in understanding the fit to human judgements, and we should instead depend on formal/mathematical measures of relationship such as Pearson's correlation or Kendall's τ_b .

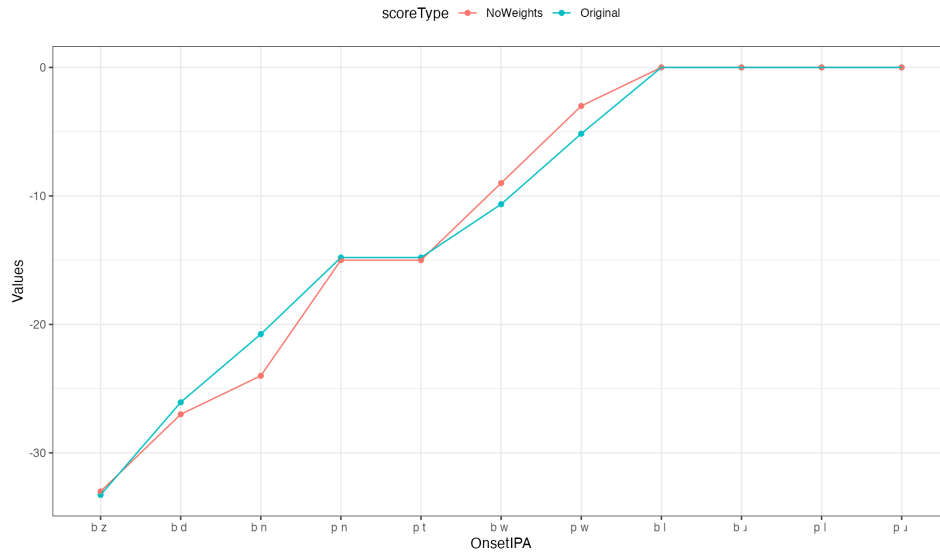


Figure 16: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the No-weights model where the latter was assigned a value of three for each constraint violation for the Albright (2009) data

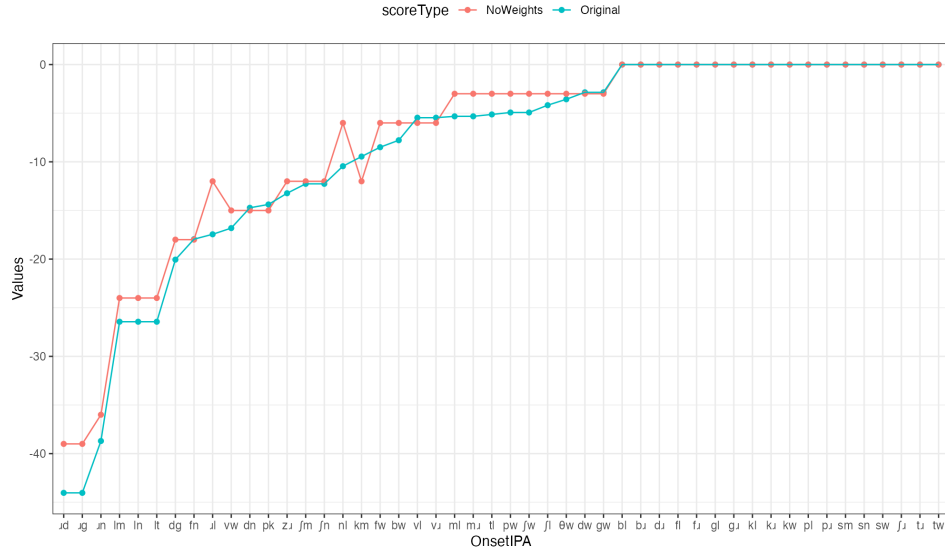


Figure 17: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the No-weights model where the latter was assigned a value of three for each constraint violation for the Daland et al. (2011) data

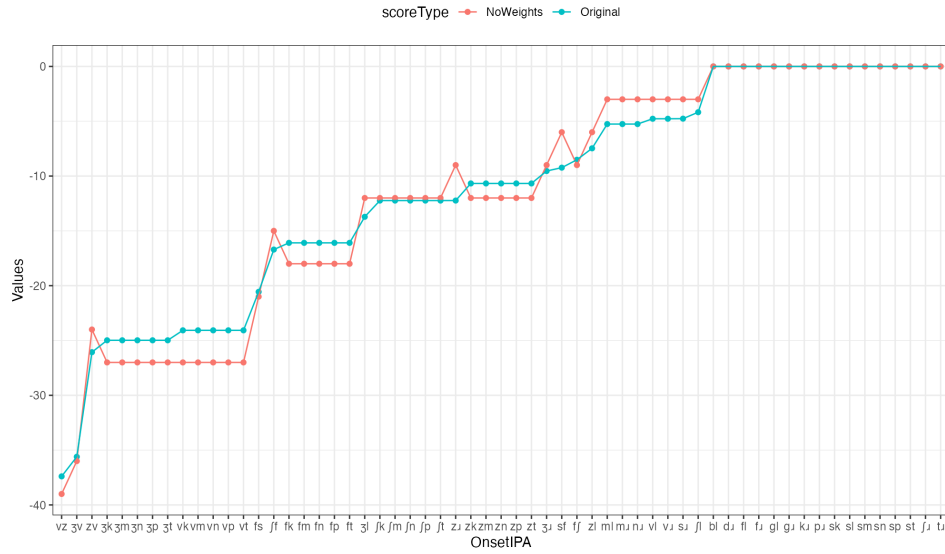


Figure 18: The harmony scores of Hayes & Wilson Phonotatic Model compared to those for the No-weights model where the latter was assigned a value of three for each constraint violation for the Scholes (1966) data

C Predicted log transformed scores for the minimal (categorical) phonotatic model

Table 14: The log transformed counts of the minimal (categorical) phonotatic model for the Albright (2009) data

IPA	log score
bl	0
bɪ	0
pl	0
pɪ	0
bw	2.197
pw	2.197
pn	2.996
bd	3.135
bn	3.135
pt	3.258
bz	4.06

Table 15: The log transformed counts of the minimal (categorical) phonotatic model for the Daland et al. (2011) data

IPA	log score
bl	0
bɪ	0
dɪ	0
dw	0
fl	0
fɪ	0
gl	0
gɪ	0
gw	0
kl	0

kı	0
kw	0
pl	0
pı	0
sm	0
sn	0
sw	0
ŷı	0
tı	0
tw	0
w	0
ŷw	0.693
tl	0.693
vl	0.693
vı	0.693
zı	0.693
ŷl	1.609
bw	2.197
pw	2.197
ŷm	2.197
ŷn	2.303
dg	2.639
dn	2.639
fw	2.773
vw	2.773
fn	2.833
km	3.135
pk	3.367
ml	3.784
mı	3.784
nl	3.784
ıl	4.043
ıg	4.22
ıd	4.29

.m	4.29
lm	4.304
ln	4.304
lt	4.344

Table 16: The log transformed counts of the minimal (categorical) phonotactic model for the Scholes (1966) data

IPA	log score
bl	0
dɪ	0
fl	0
fɪ	0
gl	0
gɪ	0
kɪ	0
pɪ	0
sk	0
sl	0
sm	0
sn	0
sp	0
st	0
ʃɪ	0
tɪ	0
sɪ	0.693
vl	0.693
vɪ	0.693
zl	0.693
zɪ	0.693
ʒɪ	1.099
ʃl	1.609
zm	1.609
zn	1.609

zk	1.792
zp	1.792
zt	1.792
zl	1.792
fm	2.197
fn	2.303
fk	2.398
fp	2.398
ft	2.485
zm	2.565
zn	2.639
zk	2.773
zp	2.773
fn	2.833
zt	2.833
vn	2.996
fm	3.045
ft	3.091
vm	3.178
fp	3.258
vt	3.258
fk	3.332
vp	3.401
vk	3.466
ml	3.784
ml	3.784
nl	3.784
sf	3.784
zv	3.829
ff	3.932
zv	3.951
vz	4.025
fs	4.078
ff	4.078

References

- Abbott, Edwin A (1884). *Flatland: A romance of many dimensions*. New York, NY, USA: Dover Thrift Edition (1992 unabridged).
- Albright, Adam (2007). *Natural classes are not enough: biased generalization in novel onset clusters*. Ms., Massachusetts Institute of Technology.
- Albright, Adam (May 2009). “Feature-based generalisation as a source of gradient acceptability.” *Phonology* 26.1, pp. 9–41.
- Albright, Adam and Bruce Hayes (2003). “Rules vs. analogy in English past tenses: a computational/experimental study.” *Cognition* 90.2, pp. 119–161.
- Alves, Fernando C (2023). “Categorical versus gradient grammar in phonotactics.” *Language and Linguistics Compass* 17.5, e12501.
- Armstrong, Sharon Lee, Lila R. Gleitman, and Henry Gleitman (1983). “What some concepts might not be.” *Cognition* 13.3, pp. 263–308.
- Bach, Emmon and R. T. Harms (1972). “How do languages get crazy rules?” *Linguistic Change and Generative Theory*. Ed. by Robert Stockwell and Ronald Macaulay. Bloomington: Indiana University Press, pp. 1–21.
- Bailey, Todd M. and Ulrike Hahn (2001). “Determinants of wordlikeness: phonotactics or lexical neighborhoods?” *Journal of Memory and Language* 44.4, pp. 568–591.
- Blevins, Juliette (1995). “The syllable in phonological theory.” *The handbook of phonological theory*. Ed. by John Goldsmith. Cambridge, MA, USA: Blackwell, pp. 206–244.
- Buckley, Eugene (2000). “On the naturalness of unnatural rules.” *UCSB Working Papers in Linguistics* 9.
- Caplan, Spencer, Jordan Kodner, and Charles Yang (2020). “Miller’s monkey updated: Communicative efficiency and the statistics of words in natural language.” *Cognition* 205, p. 104466.
- Capodici, Frank, Bruce Hayes, and Colin Wilson (2009). *Manual: Phonotactic learning program*.
- Chomsky, Noam (1965). *Aspects of the Theory of Syntax*. Cambridge: MIT Press.
- Chomsky, Noam (1986a). *Barriers*. Vol. 13. MIT press.
- Chomsky, Noam (1986b). *Knowledge of language: its nature, origin, and use*. New York, NY, USA: Praeger.
- Chomsky, Noam and Morris Halle (1968). *The Sound Pattern of English*. New York: Harper and Row.

- Coetzee, Andries W. and Joe Pater (2008). “Weighted constraints and gradient restrictions on place co-occurrence in Muna and Arabic.” *Natural Language & Linguistic Theory* 26.2, pp. 289–337.
- Cohen, Jonathan, Brian MacWhinney, Matthew Flatt, and Jefferson Provost (1993). “PsyScope: An interactive graphic system for designing and controlling experiments in the psychology laboratory using Macintosh computers.” *Behavior Research Methods, Instruments, & Computers* 25, pp. 257–271.
- Coleman, John and Janet B. Pierrehumbert (1997). “Stochastic phonological grammars and acceptability.” *Proceedings of the 3rd Meeting of the ACL Special Interest Group in Computational Phonology*. Association for Computational Linguistics, pp. 49–56.
- Cruise, Robin (2019). *Why does adding more terms into a linear model always increase the r-squared value?* Cross Validated. URL:<https://stats.stackexchange.com/q/401466> (version: 2019-04-09). eprint: <https://stats.stackexchange.com/q/401466>.
- Cumming, Geoff and Sue Finch (2005). “Inference by eye: confidence intervals and how to read pictures of data.” *American psychologist* 60.2, p. 170.
- Dai, Huteng (2025). “An exception-filtering approach to phonotactic learning.” *Phonology* 42, e5. DOI: [10.1017/S0952675725000028](https://doi.org/10.1017/S0952675725000028).
- Daland, Robert, Bruce Hayes, James White, Marc Garellek, Andrea Davis, and Ingrid Norrmann (2011). “Explaining sonority projection effects.” *Phonology* 28.2, pp. 197–234.
- Dehaene, Stanislas (2003). “The neural basis of the Weber–Fechner law: a logarithmic mental number line.” *Trends in Cognitive Sciences* 7.4, pp. 145–147. ISSN: 1364-6613. DOI: [https://doi.org/10.1016/S1364-6613\(03\)00055-X](https://doi.org/10.1016/S1364-6613(03)00055-X).
- Dehaene, Stanislas, Véronique Izard, Elizabeth Spelke, and Pierre Pica (2008). “Log or Linear? Distinct Intuitions of the Number Scale in Western and Amazonian Indigene Cultures.” *Science* 320.5880, pp. 1217–1220. DOI: [10.1126/science.1156540](https://doi.org/10.1126/science.1156540). eprint: <https://www.science.org/doi/pdf/10.1126/science.1156540>.
- Du, Naiyan and Karthik Durvasula (2025). “Psycholinguistics and Phonology: the forgotten foundations of generative phonology.” *Cambridge Elements on Phonology*.
- Duhem, Pierre Maurice Marie (1954). *The Aim and Structure of Physical Theory*. Princeton University Press.
- Durvasula, Karthik (2020). “O gradience, whence do you come?” Plenary talk at the 2020 Annual Meeting on Phonology, Santa Cruz, USA.
- Durvasula, Karthik and Adam Liter (2020). “There is a simplicity bias when generalising from ambiguous data.” *Phonology* 37.2, pp. 177–213.

- Firestone, Chaz and Brian J. Scholl (2016). “Cognition does not affect perception: Evaluating the evidence for “top-down” effects.” *Behavioral and Brain Sciences* 39, e229. DOI: [10.1017/S0140525X15000965](https://doi.org/10.1017/S0140525X15000965).
- Freyer, Frederic Jason (2017). “Emulating Language Acquisition with Stochastic Gradient Descent: A New Approach to Modeling Phonotactics.” M.A. Thesis. Brandeis University, Graduate School of Arts and Sciences.
- Friederici, Angela D and Jeanine MI Wessels (1993). “Phonotactic knowledge of word boundaries and its use in infant speech perception.” *Perception & psychophysics* 54.3, pp. 287–295.
- Frisch, Stefan A., Janet B. Pierrehumbert, and Michael B. Broe (2004). “Similarity avoidance and the OCP.” *Natural Language & Linguistic Theory* 22.1, pp. 179–228.
- Gigerenzer, Gerd (2008). “Why heuristics work.” *Perspectives on psychological science* 3.1, pp. 20–29.
- Goad, Heather (2012). “sC clusters are (almost always) coda-initial.” *The Linguistic Review* 29.3, pp. 335–373.
- Goldsmith, John Anton (1976). “Autosegmental phonology.” PhD dissertation. Massachusetts Institute of Technology.
- Goldwater, Sharon and Mark Johnson (2003). “Learning OT constraint rankings using a maximum entropy model.” *Proceedings of the workshop on variation within Optimality Theory*, pp. 111–120.
- Gorman, Kyle (2013). “Generative Phonotactics.” PhD dissertation. University of Pennsylvania.
- Greenberg, Joseph H. and James J. Jenkins (1964). “Studies in the psychological correlates of the sound system of American English.” *Word* 20.2, pp. 157–177.
- Grippando, Shannon (2019). “Sounds over Symbols? The Role of Auditory Cues in Orthographically-Related Speech Behavior.” *Languages* 4.3. ISSN: 2226-471X. DOI: [10.3390/languages4030072](https://doi.org/10.3390/languages4030072).
- Hay, J.B., J.B. Pierrehumbert, and M. E. Beckman (2003). “Speech perception, well-formedness, and the statistics of the lexicon.” Ed. by J. Local R. Ogden and R. Temple. Cambridge, U.K.: Cambridge University Press.
- Hayes, Bruce (2000). “Gradient well-formedness in Optimality Theory.” *Optimality Theory: Phonology, syntax, and acquisition*. Ed. by Joost Dekkers, Frank van der Leeuw, and Jeroen van de Weijer. Oxford: Oxford University Press, pp. 88–120.

- Hayes, Bruce and Colin Wilson (2008). “A maximum entropy model of phonotactics and phonotactic learning.” *Linguistic Inquiry* 39.3, pp. 379–440.
- Hlavac, Marek (2018). *stargazer: Well-Formatted Regression and Summary Statistics Tables*. R package version 5.2.2. Central European Labour Studies Institute (CELSI). Bratislava, Slovakia.
- Inkelas, Sharon, Aylin Küntay, Orhan Orgun, and Ronald Sprouse (2000). “Turkish electronic living lexicon (TELL).” *Turkic Languages* 4, pp. 253–275.
- Juszyk, Peter W, Angela D Friederici, Jeanine MI Wessels, Vigdis Y Svenkerud, and Ann Marie Juszyk (1993). “Infants’ sensitivity to the sound patterns of native language words.” *Journal of memory and language* 32.3, pp. 402–420.
- Kahn, Daniel (1976). “Syllable-based generalizations in English phonology.” Ph.D. Dissertation. MIT.
- Kahng, Jimin and Karthik Durvasula (2023). “Can you judge what you don’t hear? Perception as a source of gradient wordlikeness judgements.” *Glossa: a journal of general linguistics* 8.1.
- Kang, Yoonjung (2011). “Loanword Phonology.” *The Blackwell Companion to Phonology*. Ed. by Marc van Oostendorp, Colin Ewen, Elizabeth Hume, and Keren Rice. Vol. IV. Hoboken, N.J.: Wiley-Blackwell, pp. 2258–2281.
- Kaye, Jonathan, Jean Lowenstamm, and Jean-Roger Vergnaud (1985). “The internal structure of phonological elements: a theory of charm and government.” *Phonology Yearbook* 2.1, pp. 305–328. DOI: [10.1017/S0952675700000476](https://doi.org/10.1017/S0952675700000476).
- Kazanina, Nina, Jeffrey S. Bowers, and William Idsardi (2017). “Phonemes: Lexical access and beyond.” *Psychonomic Bulletin & Review*. ISSN: 1531-5320. DOI: [10.3758/s13423-017-1362-0](https://doi.org/10.3758/s13423-017-1362-0).
- Knight, Frank H (1921). *Risk, uncertainty and profit*. Boston and New York, USA: Dover 2006 unabridged republication of the edition published by Houghton Mifflin Company.
- Kostyszyn, Kalina and Jeffrey Heinz (2021). “Categorical account of gradient acceptability of word-initial Polish onsets.” *Proceedings of the Annual Meetings on Phonology*.
- Lakatos, Imre (1970). “Falsification and the Methodology of Scientific Research Programmes.” *Criticism and the Growth of Knowledge: Proceedings of the International Colloquium in the Philosophy of Science, London, 1965*. Ed. by Imre Lakatos and Alan Musgrave. Vol. 4. Cambridge University Press, pp. 91–196. DOI: [10.1017/CB09781139171434.009](https://doi.org/10.1017/CB09781139171434.009).

- Legendre, Geraldine, Yoshiro Miyata, and Paul Smolensky (1990a). "Harmonic Grammar—A formal multi-level connectionist theory of linguistic well-formedness: An application." *Proceedings of the 12th Annual Conference of the Cognitive Science Society*. Hillsdale, NJ, USA: Erlbaum, pp. 884–891.
- Legendre, Geraldine, Yoshiro Miyata, and Paul Smolensky (1990b). "Harmonic Grammar—A formal multi-level connectionist theory of linguistic well-formedness: Theoretical foundations," pp. 388–395.
- Lenneberg, Eric H (1967). "The biological foundations of language." *Hospital Practice* 2.12, pp. 59–67.
- Lewis, Harry R and Christos H Papadimitriou (1998). "Elements of the Theory of Computation."
- Linell, Per (1979). "Psychological reality in phonology. A theoretical study." *Cambridge Studies in Linguistics London* 25, pp. 1–295.
- MacWhinney, Brian (2000). *The CHILDES project: Tools for analyzing talk. Third Edition. (supported by: NICHD HD082736)*. Marwah, NJ, USA: Lawrence Erlbaum Associates.
- Mayer, Connor (in press). "Reconciling categorical and gradient models of phonotactics." *Proceedings of the Society for Computation in Linguistics*.
- Miller, George A. (1957). "Some Effects of Intermittent Silence." *The American Journal of Psychology* 70.2, pp. 311–314. ISSN: 00029556.
- Mousavi, Shabnam and Gerd Gigerenzer (2014). "Risk, uncertainty, and heuristics." *Journal of Business Research* 67.8, pp. 1671–1678.
- Ohala, John J. and Manjari Ohala (1978). "Testing hypotheses regarding the psychological manifestation of morpheme structure constraints." *Report of the Phonology Laboratory (Berkeley)* 2, pp. 156–165.
- Ohala, John J. and Manjari Ohala (1986). "Testing hypotheses regarding the psychological manifestation of morpheme structure constraints." *Experimental Phonology*. Ed. by John J. Ohala and Jeri J. Jaeger. Orlando, Florida: Academic Press, pp. 239–252.
- Persson, Anna, Santiago Barreda, and T Florian Jaeger (2025). "Comparing accounts of formant normalization against US English listeners' vowel perception." *The Journal of the Acoustical Society of America* 157.2, pp. 1458–1482.
- Pierrehumbert, Janet B. (2001). "Why phonological constraints are so coarse-grained." *Language and Cognitive Processes* 16.5–6, pp. 691–698.

- Pierrehumbert, Janet B. (2003). “Probabilistic phonology: discrimination and robustness.” *Probabilistic Linguistics*. Ed. by Rens Bod, Jennifer Hay, and Stefanie Jannedy. Cambridge: MIT Press. Chap. 6, pp. 177–228.
- Piñeiro, Gervasio, Susana Perelman, Juan P Guerschman, and José M Paruelo (2008). “How to evaluate models: observed vs. predicted or predicted vs. observed?” *Ecological modelling* 216.3-4, pp. 316–322.
- Pizzo, Presley (2015). “Investigating Properties of Phonotactic Knowledge Through Web-Based Experimentation.” Ph.D. Dissertation. University of Massachusetts, Amherst.
- Quine, Willard V. O. (1951). “Two Dogmas of Empiricism.” *Philosophical Review* 60.1, pp. 20–43. DOI: [10.2307/2266637](https://doi.org/10.2307/2266637).
- R Core Team (2021). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria.
- Rawski, Jonathan (2021). “Structure and learning in natural language.” PhD dissertation. State University of New York at Stony Brook.
- Reiss, Charles (2017). “Substance free phonology.” *The Routledge handbook of phonological theory*. Routledge, pp. 425–452.
- Sanchez, Alessandro, Stephan C Meylan, Mika Braginsky, Kyle E MacDonald, Daniel Yurovsky, and Michael C Frank (2019). “childes-db: A flexible and reproducible interface to the child language data exchange system.” *Behavior research methods* 51.4, pp. 1928–1941.
- Sarver, Isaac (2020). “A Systematic Evaluation of Computational Models of Phonotactics.” M.A. Thesis. East Lansing, MI, USA: Michigan State University.
- Scheer, Tobias (2004). *A lateral theory of phonology. Vol 1: What is CVCV, and why should it be?* Berlin: Mouton de Gruyter.
- Scheer, Tobias (2009). *Who is afraid of crazy rules*. Paper presented at Going Romance, 3–5 December, 2009. Nice.
- Scholes, Robert J. (1966). *Phonotactic Grammaticality*. The Hague: Mouton.
- Schütze, Carson T. (1996). *The Empirical Base of Linguistics: Grammaticality Judgments and Linguistic Methodology*. Chicago: University of Chicago Press.
- Schütze, Carson T. (2005). “Thinking about what we are asking speakers to do.” *Linguistic evidence: Empirical, theoretical, and computational perspectives*, pp. 457–485.
- Sebastián-Gallés, Núria and Laura Bosch (2002). “Building phonotactic knowledge in bilinguals: role of early exposure.” *Journal of Experimental Psychology: Human Perception and Performance* 28.4, p. 974.

- Shademan, Shabnam (2006). “Is phonotactic knowledge grammatical knowledge?” *Proceedings of the 25th West Coast Conference on Formal Linguistics*, pp. 371–379.
- Smith, Jennifer L. (2006). “Loan phonology is not all perception: evidence from Japanese loan doublets.” *Proceedings of the 14th Annual Japanese/Korean Linguistics Conference*, pp. 63–74.
- Smith, Jennifer L. (to appear). “Loanword Phonology.” *The Cambridge Handbook of Phonology, 2nd edition*. Ed. by Adam Jardine and Paul de Lacy. Cambridge University Press.
- Smolensky, Paul and Géraldine Legendre (2006). *The harmonic mind: From neural computation to optimality-theoretic grammar (Cognitive architecture), Vol. 1*. MIT press.
- Stoehr, Antje and Clara D. Martin (2022). “The impact of orthographic forms on speech production and perception: An artificial vowel-learning study.” *Journal of Phonetics* 94, p. 101180. ISSN: 0095-4470. DOI: <https://doi.org/10.1016/j.wocn.2022.101180>.
- Sun, John Z., Grace I. Wang, Vivek K Goyal, and Lav R. Varshney (2012). “A framework for Bayesian optimality of psychophysical laws.” *Journal of Mathematical Psychology* 56.6, pp. 495–501. ISSN: 0022-2496. DOI: <https://doi.org/10.1016/j.jmp.2012.08.002>.
- Swanson, Logan, Jeffrey Heinz, and Jonathan Rawski (accepted). “Phonotactic Learning with Structure, not Statistics.” *Linguistic Inquiry*.
- Valian, Virginia (1982). “Psycholinguistic experiment and linguistic intuition.” *Language, Mind, and Brain*. Ed. by Thomas W. Simon Simon and Robert J. Scholes. Hillsdale, NJ: Erlbaum, pp. 179–188.
- Vendelin, Inga and Sharon Peperkamp (2006). “The influence of orthography on loanword adaptations.” *Lingua* 116, pp. 996–1007.
- Vitevitch, Michael S. and Paul A. Luce (1998). “When words compete: Levels of processing in perception of spoken words.” *Psychological Science* 9.4, pp. 325–329.
- Vitevitch, Michael S. and Paul A. Luce (1999). “Probabilistic phonotactics and neighborhood density in spoken word recognition.” *Journal of Memory and Language* 40.3, pp. 374–408.
- Vitevitch, Michael S., Paul A. Luce, Jan Charles-Luce, and David Kemmerer (1997). “Phonotactics and syllable stress: implication for the processing of spoken nonsense words.” *Language and Speech* 40.1, pp. 47–62.
- “Von Neumann’s Elephant” (2025). *Wikipedia, Wikimedia Foundation*.
- Warner, Natasha, Allard Jongman, Joan Sereno, and Rachèl Kemps (2004). “Incomplete neutralization and other sub-phonemic durational differences in production and percep-

- tion: evidence from Dutch.” *Journal of Phonetics* 32.2, pp. 251–276. ISSN: 0095-4470.
DOI: [https://doi.org/10.1016/S0095-4470\(03\)00032-9](https://doi.org/10.1016/S0095-4470(03)00032-9).
- Weide, Robert L. (1994). “CMU Pronouncing Dictionary.” URL: <http://www.speech.cs.cmu.edu/cgi-bin/cmudict>.
- Wickham, Hadley (2017). *tidyverse: Easily Install and Load the ‘Tidyverse’*. R package version 1.2.1.
- Wilson, Colin and Gillian Gallagher (2018). “Accidental gaps and surface-based phonotactic learning: a case study of South Bolivian Quechua.” *Linguistic Inquiry* 49.3, pp. 610–623.
- Yang, Charles (2016). *The price of productivity*. Boston, MA, USA: MIT press.